

The Future of Growth and Human Labor Under Recursive AI Self-Improvement

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Abstract

I develop an endogenous-growth model in which a monopolistic AI developer supplies AI production services and uses recursive AI self-improvement to improve AI efficiency. I characterize the model's long-run equilibria under a finite upper bound on AI efficiency. The equilibrium regime depends jointly on the elasticity of substitution between labor and AI and on the level of this upper bound. When labor remains a production bottleneck, output per worker and wages grow at an exogenously given rate of labor-augmenting technological progress, and labor retains a positive income share. If the elasticity of substitution exceeds one and the upper bound is sufficiently high, the economy instead becomes AI-dominated. In this regime, wage growth exceeds the rate of labor-augmenting technological progress, but output per worker grows even faster. Labor's income share therefore converges to zero. With complementarity, long-run output-per-worker growth cannot exceed the rate of labor-augmenting technological progress, even without an upper bound on AI efficiency. With substitution, the long-run output-per-worker growth rate increases as AI efficiency increases without bound.

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1 Introduction

Advances in artificial intelligence raise both hopes and fears. One hope is that increasingly capable AI could generate unprecedented economic prosperity. One fear is that workers who depend on labor income could lose relative economic power. This fear, however, does not require real wages to decline: workers may receive a smaller share of a much larger economy even as their wages rise. Can unprecedented prosperity coexist with a vanishing labor income share? Under what conditions could the world economy follow such a path? How long could it take before the transition became visible, and what economic signals would reveal it? What would happen to wages and labor’s income share along the way?

These questions are becoming more pressing as AI increasingly contributes to its own development. Recursive self-improvement (RSI) is a feedback process in which better AI helps develop still better AI. OpenAI and Anthropic report that AI systems already accelerate parts of their research (OpenAI, 2026; Favaro and Clark, 2026).¹ OpenAI’s chief scientist, Pachocki (2026), writes that the pace of progress “could be sustained into recursive self-improvement.” Anthropic nevertheless cautions that, regarding AI systems that autonomously design and develop their successors, “We are not there yet” (Favaro and Clark, 2026). These reports motivate studying the economic consequences of autonomous AI research without assuming when it will become possible.

I study these questions by extending the Ramsey–Cass–Koopmans model. Competitive firms produce the final good with capital, effective labor, and AI production services. A monopolistic AI developer supplies these services and uses AI research services to improve AI efficiency, creating recursive self-improvement. Together, the elasticity of substitution between AI production services and labor and the maximum attainable level of AI efficiency determine whether human labor remains a long-run production bottleneck.

The escape from the bottleneck arises from two feedback loops. A technological loop runs from AI efficiency to AI research: greater efficiency produces more effective research services from a given amount of research compute and thereby helps create still better AI. An economic loop runs from AI efficiency to output and back to AI research: better AI raises current production and expands the resources that can finance research compute and therefore further AI improvements. Research compute, however, competes with the inference compute used in final-good production.

I first characterize the labor-bottleneck regime. It arises when the elasticity of substitution between AI production services and labor is at most one, and can also arise when the elasticity exceeds one but the maximum attainable level of AI efficiency remains below an explicit threshold. In this regime, output per worker and the real wage eventually grow at the exogenous rate of labor-augmenting technological progress, and labor’s income share converges to a positive value.

The outcome changes when the elasticity of substitution between AI production services and labor exceeds one and the maximum attainable level of AI efficiency lies above the threshold.

¹On September 6, 2026, OpenAI (2026) reported reaching its automated research-intern milestone: systems that execute well-defined research tasks under human direction. It also described progress toward an automated AI researcher by March 2028. This is a development objective, not an established arrival date for fully autonomous RSI.

An AI-dominated long-run regime then becomes possible. In this regime, capital and effective AI production services eventually grow faster than effective labor. The long-run growth rates of output per worker and the real wage exceed their labor-bottleneck values, and the interest rate is higher. Because output per person grows faster than wages, labor’s income share converges to zero. AI-driven output growth raises wages despite the shrinking fraction of output paid to labor. Within the AI-dominated regime, long-run growth and the interest rate increase with the maximum attainable level of AI efficiency.

I also examine the role of the upper bound on AI efficiency. With complementarity, long-run output-per-person growth cannot exceed the exogenous rate of labor-augmenting technological progress, even without an upper bound. At unit elasticity and without an upper bound, I derive conditions for balanced growth in which output per person and real wages grow faster than this exogenous rate. With substitution, long-run growth and the interest rate in the AI-dominated equilibria increase without bound as the efficiency upper bound rises.

Simulations show that the AI-dominated long-run regime need not become visible quickly. The economy may remain on a moderate-growth path for a long time. Rapid growth of effective AI production services and an interest rate above the labor-bottleneck benchmark can appear before the output and wages start to accelerate. When the economy begins far from its steady state, the transition could also require a large initial research outlay relative to AI-industry revenue and temporary developer losses that are offset in present value by future monopoly rents.² Alternative calibrations produce fast or slow transitions.

This paper has seven sections, including the present introduction. Section 2 relates the paper to the literature. Section 3 presents the model and defines equilibrium. Section 4 establishes sufficient conditions for equilibrium with a finite upper bound on AI efficiency and characterizes the associated long-run growth and distributional outcomes. Section 5 examines the three substitution regimes without an upper bound on AI efficiency. Section 6 presents some equilibrium numerical simulations. Section 7 concludes. The appendix contains the proofs and describes the algorithm used for the simulations.

2 Related literature

AI, growth, and automated research. My RSI technology builds on the endogenous-growth tradition, in which research investment produces nonrival knowledge (Romer, 1990; Aghion and Howitt, 1992; Jones, 1995). Recent work asks how AI affects the economy when AI can automate production, research, or both. Trammell and Korinek (2023) distinguish the automation of ordinary production from the automation of R&D. Mullainathan and Rambachan (2025) argue that AI may also reorganize science by bringing idea generation and theory selection into algorithmic workflows. Endogenous AI development has direct antecedents: Lu (2021) models AI as a nonrival input that

²The initial-loss example is the full-price-decline calibration retained in the replication files, not the two comparisons displayed in Section 6. Section 6.4 explains its limitations.

can contribute to its own accumulation, with resources allocated to its development. [Gersbach et al. \(2025\)](#) model AI as self-learning capital that improves through application, with knowledge spillovers and skilled labor allocated between AI and applied research.

[Davidson et al. \(2026\)](#) provide a closely related feedback architecture: better AI makes further AI research more productive, while higher output expands the resources available for research. Their aggregate-economy application holds the saving rate and resource-allocation shares fixed. GATE ([Erdil et al., 2025](#)) instead optimizes investment and the allocation of compute between training and inference through a social planner’s problem, with adjustment costs for compute and conventional capital. These distinctions matter: endogenous technological progress need not imply a private, profit-maximizing choice of research spending. In my model, the developer chooses research compute by comparing its current cost with the value of future monopoly profit. This decision determines the pace of AI improvement jointly with household saving, capital accumulation, inference demand, and equilibrium prices. Research productivity remains a technological assumption; the resources committed to exploiting it are an intertemporal economic choice.

This mechanism also differs from imposing a path of AI progress. In their baseline model, [Korinek and Suh \(2024\)](#) study how an exogenous automation frontier interacts with capital accumulation during a possible transition to AGI. [Korinek et al. \(2026\)](#) map alternative paths of AI capabilities and adoption into GDP, wages, factor shares, and occupational outcomes through 2030, embedding recursive improvement in the assumed path of AI task productivity.

Substitution, bottlenecks, and long-run growth. Following [Davidson et al. \(2026\)](#), I call an input or task a bottleneck when its slow improvement constrains aggregate progress even if other inputs or tasks improve rapidly. [Aghion et al. \(2019\)](#) apply Baumol’s cost-disease logic to AI and growth: if some essential production tasks remain difficult to automate, progress in automatable tasks need not translate into comparably rapid aggregate growth. [Jones \(2025\)](#) develops the corresponding mechanism inside R&D. The effect of AI on research depends jointly on the share of research tasks AI can perform, its productivity in those tasks, and the strength of complementarities across tasks. Tasks that AI cannot perform can therefore limit overall research progress. In my model, human labor no longer enters AI research, but it may remain essential in final-good production.

The production side connects the CES growth tradition ([Pitchford, 1960](#)) to models in which accumulable inputs can sustain growth without a vanishing marginal product ([Jones and Manuelli, 1990](#); [Rebelo, 1991](#)). [Duffy and Papageorgiou \(2000\)](#) emphasize that a high elasticity is not by itself sufficient: the productivity of the accumulable input must also be high enough. In the AI literature, [Nordhaus \(2021\)](#) identifies substitutability between information and conventional inputs as central to an economic singularity. [Jones and Tonetti \(2026\)](#) endogenize automation in goods and idea production, treating AI in one scenario as a continuation of historical automation. Complementary tasks create weak links that delay growth acceleration. Consistent with this mechanism, [Demirer et al. \(2026\)](#) find that large AI-induced gains in coding activity attenuate sharply when measured by

software releases, and estimate strong complementarity between upstream output and downstream human effort. Finally, Restrepo (2025) distinguishes work that remains a bottleneck from work that is no longer essential for growth. I obtain a related distinction without assuming that all work can be performed by AI. Gross substitution between AI services and labor is necessary but not sufficient for labor to cease being a bottleneck: attainable AI efficiency must also be high enough. I derive the threshold separating the labor-bottleneck and AI-dominated equilibria. The capability threshold in Gans (2025) concerns a different mechanism—whether AI directs research toward dense or exploratory regions of the knowledge space—rather than whether effective AI services can replace labor in final production.

Adoption and transition dynamics. Technological availability need not translate immediately into measured productivity gains. Brynjolfsson et al. (2021) study complementary intangible investment and show how its imperfect measurement generates a productivity J-curve: measured productivity growth initially understates the gains and later overstates them. Information also matters. Beaudry and Portier (2014) review how news about future technology changes current economic decisions. These mechanisms distinguish technological progress, its implementation, and the response to learning about it. I do not model organizational capital or compute-adjustment costs. My RSI-activation exercises isolate an unanticipated change in research opportunities, allowing consumption and research spending to adjust immediately.

Wages and the labor income share. The distributional results connect to the task approach to automation. Acemoglu and Restrepo (2019) distinguish the displacement effect of automating existing tasks from two forces that can increase labor demand: the expansion of output generated by higher productivity and the creation of new tasks in which labor retains a comparative advantage, which they call the reinstatement effect. My model does not explicitly model the automation or creation of tasks. It instead studies how substitution between labor and AI services interacts with productivity gains. Labor and AI services enter a constant-elasticity-of-substitution (CES) production function. Holding capital and effective labor fixed, greater use of AI services raises labor’s income share when the elasticity is below one, leaves it unchanged when the elasticity equals one, and reduces it when the elasticity exceeds one. A related CES mechanism underlies Karabarbounis and Neiman (2014), who study how a decline in the relative price of investment induces substitution from labor toward capital and lowers labor’s income share. Recent work shows that automation can raise wages while reducing labor’s income share (Autor and Kausik, 2026). In the long run, Ray and Mookherjee (2022) show that capital accumulation can drive labor’s share to zero while real wages rise indefinitely, and Restrepo (2025) obtains a vanishing labor share when compute can perform all bottleneck work. The coexistence of rising real wages and a declining labor share is therefore not itself new. In Farboodi et al. (2026), data generated by economic activity improve automated tasks and extend automation to new tasks. With endogenous capital accumulation, this mechanism can generate explosive growth but stagnant long-run wages. My contribution is a sharper equilibrium characterization. Among the long-run regimes characterized

here with a finite upper bound on AI efficiency, real-wage growth exceeds the exogenous rate only in the AI-dominated regime, in which labor’s income share converges to zero. In every positive-labor-share regime, real-wage growth equals the exogenous rate. Within the AI-dominated regime, an exact expression links the wage-growth premium to the rate at which labor’s income share declines, while the AI-efficiency threshold determines which regime arises. I do not study taxation, but a vanishing labor income share connects to the erosion of the labor-income tax base relative to output analyzed by [Korinek and Lockwood \(2026\)](#). Their additional concern about a shrinking human-consumption tax base is not implied by my labor-share result.

The aggregation of labor also matters for interpreting these results. [Korinek et al. \(2026\)](#) show that average wages can rise while wages for cognitive workers fall and their unemployment increases. My representative-worker model cannot characterize this heterogeneity within labor. It instead isolates the conditions under which aggregate real wages rise faster than labor-augmenting technological progress while labor’s income share converges to zero.

Interest rates. AI-driven growth can also affect equilibrium returns. Faster expected consumption growth raises long-term real rates through consumption smoothing in [Chow et al. \(2026\)](#). The AI scenario in [Rachel \(2025\)](#) can also raise the natural rate through saving and investment. In the ICT-investment scenarios of [Melina and Villa \(2025\)](#), the short-run natural rate rises more when ICT capital complements labor; greater substitutability can weaken or reverse that response. Using recent AI-sector investment, [Wachter and Wachter \(2026\)](#) infer scenarios with rapid output and AI-sector expansion and a higher risk-free rate. I study the long-run net return on physical capital in a closed-economy Ramsey equilibrium, rather than the short-run natural rate. The return remains tied to labor-augmenting technological progress when labor is the bottleneck, but rises with attainable AI efficiency in the AI-dominated regime.

Measurement of AI services. The distinction between raw compute and the effective services it produces is informed by [Korinek and McKelvey \(2026\)](#), who distinguish physical computing capacity from quality-adjusted inference and R&D/training output and account for algorithmic progress. I retain separate compute flows for inference and research but summarize the technology that augments both in one AI-efficiency stock. [Coyle and Poquiz \(2025\)](#) document the broader difficulties of measuring AI investment and output, quality change, and changes in production processes. These measurement problems caution against interpreting the model’s AI-efficiency stock as a directly observed statistical series.

Market structure. Private incentives to automate also have important precedents. [Acemoglu and Restrepo \(2018\)](#) endogenize the allocation of a fixed supply of scientists between automation and new tasks through expected innovation profits. [Hémous and Olsen \(2022\)](#) model profit-driven automation and the creation of new products, with rising low-skill wages strengthening incentives to automate. Thus, private research incentives are not themselves new. Here they govern improvements in the efficiency of AI that is used in both production and further research.

The integrated developer is a tractable limiting case of the concentration forces studied by [Korinek and Vipra \(2025\)](#) and the AI market-power questions surveyed by [Gans \(2024\)](#). [Athey and Scott Morton \(2025\)](#) model AI as a priced intermediate input and show how upstream market power affects adoption, downstream prices, factor returns, and welfare. [Liu and Wan \(2026\)](#) compare decentralized, planner, and monopoly allocations in an endogenous-growth model linking general-purpose AI to specialized applications. They likewise connect AI industry structure to innovation incentives. Evidence from API markets on model pricing, entry, and differentiation in [Demirer et al. \(2025\)](#) also cautions against treating current AI markets as a literal monopoly. I hold market structure fixed to isolate the dynamic mechanism. This accounting distinguishes the AI industry’s gross revenue from monopoly profit: inference and research expenditure are paid before the developer distributes the residual to its owner.

Contribution. The individual mechanisms studied here—automated research, production bottlenecks, declining labor shares, rising real wages, and changes in equilibrium returns—appear in the existing literature. I combine them in a decentralized endogenous-growth model with recursive AI improvement and a profit-maximizing AI developer. The pace of AI improvement follows from the developer’s research decision, financed against future monopoly profit, rather than an imposed technological path or a fixed research share. This connects the transition in wages and income shares to the incentives and resources supporting AI development. Across the long-run equilibria characterized here with a finite upper bound on AI efficiency, real wages grow faster than exogenous labor-augmenting technological progress if and only if labor’s income share converges to zero. Output per person then grows even faster, and an exact expression links the wage-growth premium to the rate at which labor’s income share declines. Within the AI-dominated regime, higher attainable AI efficiency raises both long-run growth and the net return on capital.

3 Model

Time is continuous. The economy contains a representative household, competitive final-good firms, and an integrated AI developer. Labor-augmenting technology A grows exogenously. AI efficiency B improves through autonomous AI research: the only current input used to raise B is research compute. The model therefore describes an economy in which there are no longer human researchers developing AI and AI has achieved recursive self-improvement (RSI).

3.1 Population and labor-augmenting technology

Population grows at the exogenous rate $n \geq 0$:

$$\dot{N} = nN, \quad n \geq 0. \tag{1}$$

Each person supplies one unit of homogeneous labor inelastically. The aggregate labor endowment is therefore N . Because autonomous research uses no human labor, all labor is employed in final

production and equilibrium will impose $L = N$.

The variable A measures the level of labor-augmenting technology. One unit of labor supplies A efficiency units, so effective labor is AL . Labor-augmenting technology evolves according to

$$\dot{A} = \gamma A, \quad \gamma > 0. \quad (2)$$

3.2 Household

The representative household owns physical capital and the AI developer. Let Π denote the AI developer's net profit, which accrues to the household. The household takes the paths of the net interest rate r , the wage w and Π as given. For a predetermined capital stock $K_0 > 0$, it chooses aggregate consumption C to solve

$$\max_{\{C_t\}_{t \geq 0}} \int_0^\infty e^{-\rho t} N_t \log(C_t/N_t) dt, \quad \rho > n, \quad (3)$$

subject to

$$\dot{K} = rK + wN + \Pi - C. \quad (4)$$

The variable r is the net real interest rate: depreciation has already been deducted from the rental return to capital. The household's Euler condition is

$$\frac{\dot{C}}{C} = n + r - \rho. \quad (5)$$

Its transversality condition is

$$\lim_{t \rightarrow \infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0. \quad (6)$$

3.3 Final-good production

Competitive firms produce the final good from physical capital K , effective labor AL , and effective AI production services X . Let $0 < \alpha < 1$ denote the output elasticity with respect to capital, $\delta \geq 0$ the depreciation rate, and $\sigma > 0$ the elasticity of substitution between effective labor and effective AI production services. The parameters ω_L and ω_X denote the CES weights on effective labor and effective AI production services, respectively. Production is

$$Y = K^\alpha Z^{1-\alpha}, \quad (7)$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}. \quad (8)$$

Jobs comprise multiple tasks, and AI may substitute for human input in some tasks while complementing it in others (Autor, 2013; Acemoglu and Restrepo, 2019). I abstract from differences across workers, occupations, and tasks by modeling homogeneous labor and aggregate effective AI production services. The elasticity of substitution governs the relationship between these aggregate

inputs, rather than the allocation of individual tasks to humans or AI. This simplification focuses the analysis on aggregate wages and labor's income share, leaving differences in exposure and earnings across workers outside the model.

Let p_X denote the price of one unit of effective AI production services. Taking the input prices $r + \delta$, w , and p_X as given, a final-good firm solves

$$\max_{K, L, X \geq 0} \{K^\alpha Z(AL, X)^{1-\alpha} - (r + \delta)K - wL - p_X X\}. \quad (9)$$

Define s_X as the elasticity of Z with respect to X :

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}. \quad (10)$$

The first-order conditions are

$$r = \alpha \frac{Y}{K} - \delta, \quad (11)$$

$$w = (1 - \alpha)(1 - s_X) \frac{Y}{L}, \quad (12)$$

$$p_X = (1 - \alpha)s_X \frac{Y}{X}. \quad (13)$$

Constant returns and these conditions imply

$$Y = (r + \delta)K + wL + p_X X. \quad (14)$$

Solving the final-good firm's AI condition for its price yields the inverse demand schedule $p_X(X)$. The inverse absolute elasticity is

$$e_X \equiv -\frac{\partial \ln p_X(X)}{\partial \ln X} = \frac{1 - s_X}{\sigma} + \alpha s_X. \quad (15)$$

This demand schedule is an input to the developer's problem.

3.4 The AI developer

3.4.1 Effective AI production and research services

I use *compute* to denote the computational resources used to execute the numerical operations required by AI systems (Sastry et al., 2024). The variables U and M are rates of compute use rather than stocks of processors or data centers. I measure compute in units such that its price in final goods is one. Compute is therefore a rival, costly input, not a measure of AI performance.

I distinguish compute used to deploy AI from compute used to improve it. Inference compute U runs existing AI systems and produces outputs used in current final-good production. Research compute M is allocated to the development of better AI. It includes training and fine-tuning, but also computational experimentation, model evaluation, and other compute-intensive components

of AI research. I therefore use the term *research compute* rather than the narrower term *training compute*. Korinek and McKelvey (2026) make a related distinction between compute used for inference and compute used for R&D and training.

Raw compute does not by itself determine the quantity of AI services produced. The variable B measures AI efficiency: AI services obtained per unit of compute. Thus U and M are raw computational inputs, while BU and BM are the corresponding flows of AI production and AI research services. This interpretation parallels measures of algorithmic efficiency based on the decline in compute required to achieve a given level of performance (Hernandez and Brown, 2020; Erdil and Besiroglu, 2022). A rise in B therefore represents an improvement in the algorithms that use compute, rather than an expansion of physical computing capacity. I assume that the same B augments compute in both activities.

Inference compute produces AI production services according to

$$X = BU. \quad (16)$$

Equation (16) maps inference compute into the flow $X = BU$ of AI production services. Research compute analogously produces BM , the flow of AI research services.

Figure 1 summarizes the production block.

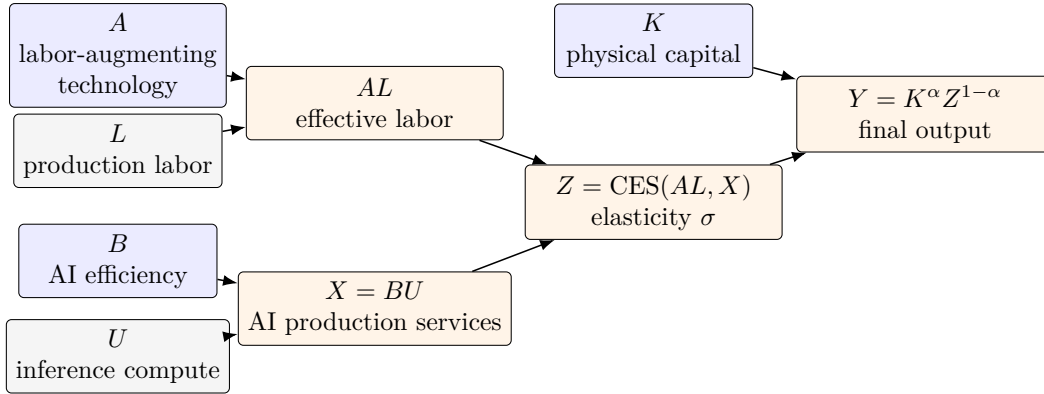


Figure 1: Production of final output and AI production services. AI efficiency B augments inference compute U to produce $X = BU$, the flow of AI production services. These services combine with effective labor AL to produce Z , which combines with capital K to produce Y .

3.4.2 RSI technology

Given initial AI efficiency $B_0 > 0$ and an upper bound $\bar{B} \in (B_0, \infty]$, AI efficiency evolves according to the RSI technology

$$\dot{B} = \chi(BM)^\eta \psi(B), \quad \psi(B) = 1 - \frac{B}{\bar{B}}, \quad \chi > 0, \quad 0 < \eta < 1. \quad (17)$$

Figure 2 summarizes the RSI mechanism.

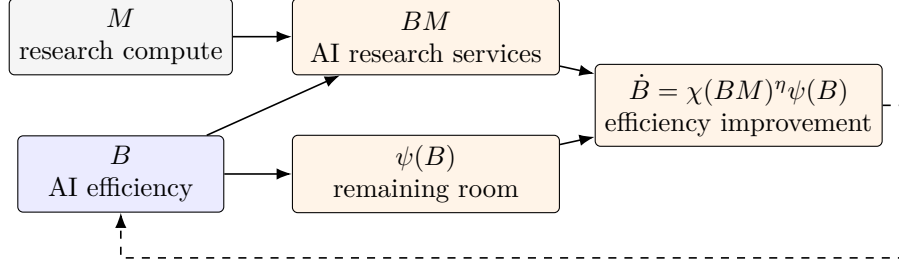


Figure 2: Recursive self-improvement in AI efficiency. AI efficiency B augments research compute M to produce BM , the flow of AI research services. These services and the remaining room for improvement $\psi(B)$ determine $\dot{B}$, which increases B .

The parameter χ shifts research productivity. The parameter η governs returns to current research scale. The same AI-efficiency index B augments both inference and research compute. This common-productivity assumption is parsimonious but substantive. Equation (17) captures RSI: a higher B produces more AI research services BM from a given amount of research compute M , and these services help improve B further. This is the AI-research feedback studied by Davidson et al. (2026). The term $\psi(B)$ reduces the remaining room for improvement as AI efficiency approaches $\bar{B}$. The uncapped specification is recovered at $\bar{B} = \infty$.

Evidence that ideas become harder to find motivates allowing research productivity to decline as technology advances. Bloom et al. (2020) document that research effort has risen substantially while research productivity has fallen across several industries, products, and firms. Following this interpretation, Davidson et al. (2026) use their estimates to discipline diminishing returns in idea production. This evidence motivates diminishing returns, but it does not establish a literal finite technological ceiling. The term $1 - B/\bar{B}$ is a deliberately stronger reduced-form specification to make sure there is an equilibrium when AI and labor are substitutes.

The upper bound on AI efficiency $\bar{B}$ allows me to establish and compute equilibria for finite values of $\bar{B}$ and then ask whether they have a meaningful limit as $\bar{B} \rightarrow \infty$. Without the upper bound, recursive improvement may instead produce a finite-time singularity: one or more endogenous quantities—such as AI efficiency, output, capital, or research compute—become unbounded at a finite date, so the trajectory cannot be continued as a path of finite allocations.

3.4.3 Supply of effective AI production services

An integrated AI developer has exclusive ownership of B . It conducts research and sells effective AI production services. The developer sells X and purchases the $U = X/B$ units of inference compute needed to supply those services. Because X does not enter the RSI technology in equation (17) directly, its within-date choice can be solved conditional on B , K , and L . Define optimized operating profit before research expenditure as

$$\pi(B) = \max_{X \geq 0} \left\{ p_X(X)X - \frac{X}{B} \right\}. \quad (18)$$

The notation $\pi(B)$ leaves its dependence on current K , A , and L implicit. The first-order condition equates marginal revenue to the compute cost of one additional service unit:

$$p_X(1 - e_X) = \frac{1}{B}. \quad (19)$$

Lemma 1 in Appendix A shows that problem (18) has a unique, strictly positive global maximizer characterized by condition (19). Thus, the developer charges the unit cost $1/B$ multiplied by the monopoly markup $1/(1 - e_X)$. The envelope theorem gives

$$\pi_B = \frac{X}{B^2} = \frac{U}{B}. \quad (20)$$

At a fixed X , a marginal increase in AI efficiency saves U/B units of the consumption good.

AI efficiency is nonrival in use. Once B has been developed, using it to augment one unit of inference or research compute does not reduce its ability to augment other units. This does not make effective AI services costless: BU and BM require the rival and costly compute inputs U and M . Nonrivalry also does not rule out excludability. An owner can restrict access to model weights, software, or AI services.

To see the resulting appropriability problem, suppose that competitive service providers had free access to B . Free entry would imply no operating rent. A developer that paid for M would improve a common AI-efficiency index B that competing providers could use without compensating it. The developer would bear the research cost but could not appropriate the resulting benefits. Exclusive ownership allows the developer to capture part of the value created by a higher B and makes the return to research appropriable, as in the standard endogenous-growth treatment of nonrival ideas (Romer, 1990).

That being said, monopoly is an institutional assumption, not a technological implication of nonrivalry. Licensing, oligopoly, public funding, or open-source provision with research subsidies could also support investment in B . Each alternative would require an additional institution and would change prices, profits, and the distribution of rents. Recent work studies these alternatives in more detail (Korinek and Vipra, 2025; Gans, 2024; Demirer et al., 2025).

3.4.4 The dynamic problem

The AI developer uses M for research and owns B and any resulting improvements in AI efficiency. After the within-date service choice, the developer's net profit is $\Pi_t = \pi(B_t) - M_t$: operating profit less research expenditure. Taking the paths of r , K , A , and L as given and substituting the static solution to problem (18), the developer chooses the path of research compute M to increase AI efficiency B :

$$\max_{\{M_t\}_{t \geq 0}} \int_0^\infty \exp \left\{ - \int_0^t r_s ds \right\} [\pi(B_t) - M_t] dt, \quad (21)$$

subject to (17) and B_0 given. Admissible research paths have $M_t \geq 0$ and finite total expenditure on every finite interval. The objective is the present value of net profit, not profit at a single date.

Let q be the current-value shadow price of AI efficiency, measured in units of the current final good per unit of B . An interior choice of research compute M satisfies

$$q\chi\eta B(BM)^{\eta-1}\psi(B) = 1, \quad (22)$$

Equation (22) equates the marginal benefit of research compute to its unit cost. Holding current B fixed, an additional unit of M raises the rate of improvement in AI efficiency by $\chi\eta B(BM)^{\eta-1}\psi(B)$. Multiplication by q values the resulting efficiency gain, accounting for future profit gains and further improvements in AI efficiency.

The current-value costate equation is

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta \frac{\dot{B}}{B} - \dot{B} \frac{\psi'(B)}{\psi(B)}. \quad (23)$$

Rearranging gives the return equation

$$rq = \frac{U}{B} + q\eta \frac{\dot{B}}{B} + q\dot{B} \frac{\psi'(B)}{\psi(B)} + \dot{q}. \quad (24)$$

The first term on the right-hand side of Equation (24) is the inference-compute saving from greater AI efficiency. The second is the research-productivity gain from RSI. With a finite upper bound, the third term is negative because a higher B leaves less room for further improvement. The final term is the capital gain on AI efficiency. Together these returns must equal the market return rq .

The developer's transversality condition is

$$\lim_{t \rightarrow \infty} \exp \left\{ - \int_0^t r_s ds \right\} q_t B_t = 0. \quad (25)$$

It rules out a path that leaves positive discounted value in the AI-efficiency state.

The preceding conditions are necessary for an interior optimum but do not establish global optimality. Lemma 2 in Appendix A gives sufficient conditions for a research path to maximize the present value of the developer's net profit in problem (21).

3.5 Equilibrium

I can now combine the decisions of the three agents and the market clearing conditions to define an equilibrium. Capital-market clearing identifies household capital with the capital rented by final-good firms. Labor-market clearing gives $L = N$. AI-service market clearing identifies the developer's sales with the final sector's purchases. Combining the household budget, final-good zero profit, developer profit, and market clearing yields the aggregate resource constraint

$$C + \dot{K} + \delta K + U + M = Y. \quad (26)$$

Definition 1 (Equilibrium). *For exogenous initial values $A_0, N_0, K_0, B_0 > 0$, an equilibrium is an*

infinite-horizon trajectory

$$\{K_t, B_t, C_t, q_t, L_t, U_t, M_t, X_t, Z_t, Y_t, r_t, w_t, p_{X,t}, s_{X,t}, e_{X,t}, \Pi_t\}_{t \geq 0} \quad (27)$$

such that for every date, the following conditions hold:

1. The intratemporal allocation and prices satisfy

$$L = N, \quad (28a)$$

$$Z = \left[\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}, \quad (28b)$$

$$Y = K^\alpha Z^{1-\alpha}, \quad (28c)$$

$$s_X = \frac{\omega_X X^{(\sigma-1)/\sigma}}{\omega_L (AL)^{(\sigma-1)/\sigma} + \omega_X X^{(\sigma-1)/\sigma}}, \quad (28d)$$

$$e_X = \frac{1 - s_X}{\sigma} + \alpha s_X, \quad (28e)$$

$$r = \alpha \frac{Y}{K} - \delta, \quad (28f)$$

$$w = (1 - \alpha)(1 - s_X) \frac{Y}{L}, \quad (28g)$$

$$p_X = (1 - \alpha) s_X \frac{Y}{X}, \quad (28h)$$

$$\frac{1}{B} = p_X (1 - e_X), \quad (28i)$$

$$X = BU, \quad (28j)$$

$$1 = q\chi\eta B(BM)^{\eta-1}\psi(B), \quad (28k)$$

$$\Pi = p_X X - U - M. \quad (28l)$$

2. The four dynamic equations are

$$\dot{K} = Y - C - U - M - \delta K, \quad (29a)$$

$$\dot{B} = \chi(BM)^\eta \psi(B), \quad (29b)$$

$$\frac{\dot{C}}{C} = n + r - \rho, \quad (29c)$$

$$\frac{\dot{q}}{q} = r - \frac{X}{qB^2} - \eta \frac{\dot{B}}{B} - \dot{B} \frac{\psi'(B)}{\psi(B)}. \quad (29d)$$

3. The two transversality conditions are

$$\lim_{t \rightarrow \infty} e^{-\rho t} \frac{N_t K_t}{C_t} = 0, \quad (30a)$$

$$\lim_{t \rightarrow \infty} \exp \left\{ - \int_0^t r_s ds \right\} q_t B_t = 0. \quad (30b)$$

Remark 1 (The economy without AI). With $\omega_X = 0$ and $\omega_L = 1$, the model reduces to the Ramsey–Cass–Koopmans economy with logarithmic utility and exogenous labor-augmenting technological progress.

The system contains no separate household budget or final-good zero-profit equation because these are redundant after market clearing. The consolidated income decomposition follows from the equations in the definition:

$$1 = \frac{(r + \delta)K}{Y} + \frac{wL}{Y} + \frac{\Pi}{Y} + \frac{U}{Y} + \frac{M}{Y}. \quad (31)$$

It records how final output is divided among gross capital income, labor income, developer profit, inference compute, and research compute. It is an accounting consequence of equilibrium, not an additional equilibrium condition.

4 Equilibrium with a finite upper bound on AI efficiency

In this section, I characterize the long-run equilibrium behavior for various parameter configurations. The key finding is that long-run output-per-worker growth can exceed the rate of labor-augmenting technological change when the elasticity of substitution between AI production services and labor exceeds one and the upper bound on AI efficiency is sufficiently high.

To identify this threshold, I calculate the net return on capital implied by the production and pricing conditions as AI efficiency approaches its upper bound and labor’s income share tends to zero. I compare this return with $\rho + \gamma$, the interest rate that sustains consumption-per-person growth at the exogenous rate γ . When AI and labor are substitutes and the former return exceeds the latter, the model supports equilibrium paths along which capital and AI production services grow faster than effective labor, even though AI efficiency converges to a finite limit.

4.1 The return to capital and the efficiency threshold

A finite upper bound restricts AI efficiency, but not the compute used to supply AI production services. To identify the return when labor’s income share vanishes, fix $B > 0$ and consider $s_X \rightarrow 1$. Since $wL/Y = (1 - \alpha)(1 - s_X)$, this is the zero-labor-share limit.

Combine the final-good firm’s demand for AI services in equation (13) with the developer’s pricing condition in equation (19). The result is:

$$\frac{X}{Y} = B(1 - \alpha)s_X(1 - e_X) \longrightarrow B(1 - \alpha)^2. \quad (32)$$

The limit uses $e_X \rightarrow \alpha$ from Equation (15). The CES aggregator in equation (8) and the definition of s_X in equation (10) imply

$$\frac{Z}{X} = \left(\frac{\omega_X}{s_X} \right)^{\sigma/(\sigma-1)} \longrightarrow \omega_X^{\sigma/(\sigma-1)}. \quad (33)$$

Dividing the production function in equation (7) by Y gives $1 = (K/Y)^\alpha (Z/Y)^{1-\alpha}$. Using $Z/Y = (Z/X)(X/Y)$ and the limits in equations (32) and (33), I obtain

$$\frac{Y}{K} = \left(\frac{Z}{Y} \right)^{(1-\alpha)/\alpha} \longrightarrow \left[(1-\alpha)^2 B \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha}. \quad (34)$$

Substituting the limiting output–capital ratio in equation (34) into the capital–return condition $r = \alpha Y/K - \delta$, equation (11), gives

$$\mathcal{R}(B) \equiv \alpha \left[(1-\alpha)^2 B \omega_X^{\sigma/(\sigma-1)} \right]^{(1-\alpha)/\alpha} - \delta. \quad (35)$$

This is a return at a particular limit of the production and pricing conditions, not necessarily an equilibrium interest rate. For $\sigma > 1$, $s_X \rightarrow 1$ means that AI production services become arbitrarily large relative to effective labor. For $\sigma < 1$, $s_X \rightarrow 1$ means that effective labor services become arbitrarily large relative to AI production services. Whether the equilibrium interest rate converges to $\mathcal{R}(\bar{B})$ depends on both σ and $\bar{B}$, as shown below.

By the Euler equation (5), consumption per person grows at $r - \rho$. A long-run equilibrium in which consumption per person grows at γ therefore requires $r = \rho + \gamma$. Define $\mathcal{B}(\sigma)$ as the AI-efficiency level at which the zero-labor-share return equals $\rho + \gamma$, the return required for consumption per person to grow at γ :

$$\begin{aligned} \mathcal{R}(\mathcal{B}(\sigma)) &= \rho + \gamma, \\ \mathcal{B}(\sigma) &= \frac{[(\rho + \gamma + \delta)/\alpha]^{\alpha/(1-\alpha)}}{(1-\alpha)^2 \omega_X^{\sigma/(\sigma-1)}}. \end{aligned} \quad (36)$$

The inequalities between $\bar{B}$ and $\mathcal{B}(\sigma)$ –whether AI can attain sufficient efficiency– and between σ and 1 –whether labor and AI are substitutes or complements– distinguish the long-run equilibrium regimes characterized below.

4.2 Equilibrium existence and long-run regimes

Define $g_{Y/N}$ as the growth rate of output per worker and g_w as the real-wage growth rate. The next proposition characterizes the long-run behavior of equilibrium trajectories.

Proposition 1 (Equilibrium existence and long-run characterization). *Fix $0 < \bar{B} < \infty$. For each parameter configuration covered by a case below, there exists a nonempty open set of initial conditions (A_0, N_0, K_0, B_0) such that every initial state in that set has an equilibrium trajectory with the following properties:*

(i) **Complementarity.** *If $0 < \sigma < 1$ and $\bar{B} > \mathcal{B}(\sigma)$, then*

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \lim_{t \rightarrow \infty} s_X \in (0, 1). \quad (37)$$

(ii) **Unit elasticity.** If $\sigma = 1$, then

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad s_X = \omega_X \quad (38)$$

(iii) **Substitution with a labor bottleneck.** If $\sigma > 1$ and $\bar{B} < \mathcal{B}(\sigma)$, then

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad \lim_{t \rightarrow \infty} s_X \in (0, 1). \quad (39)$$

(iv) **AI-dominated growth.** If $\sigma > 1$ and $\bar{B} > \mathcal{B}(\sigma)$, then

$$g_{Y/N} \rightarrow \mathcal{R}(\bar{B}) - \rho > \gamma, \quad (40)$$

$$g_w \rightarrow \gamma + \frac{\mathcal{R}(\bar{B}) - \rho - \gamma}{\sigma} > \gamma, \quad (41)$$

$$r \rightarrow \mathcal{R}(\bar{B}), \quad s_X \rightarrow 1.$$

Proof. See Appendix A, [Proof of Proposition 1](#). □

The omission of the case where $\sigma < 1$ and $\bar{B} < \mathcal{B}(\sigma)$ is deliberate as the upper bound $\bar{B}$ on AI efficiency is a device to avoid a singularity when $\sigma > 1$. In this region, any equilibrium has an AI production bottleneck, and long-run output-per-worker growth is bounded strictly below γ . See Appendix A, [Proof of the AI-bottleneck statement](#).

The proof of Proposition 1 first normalizes growing quantities and shrinking efficiency gaps. For each parameter configuration covered by the proposition, I conjecture stationary values of these normalized variables. I then construct trajectories from every sufficiently nearby initial state, deriving equilibrium trajectories and checking they converge to the stationary conjectures. Local uniqueness results are given in the proof. Existence from arbitrary initial states and global uniqueness are not established. These analytical results guide the algorithm's boundary conditions (Appendix B).

Remark 2 (Wages and AI revenue in the AI-dominated regime). In the AI-dominated equilibria of Proposition 1(iv), output per worker grows asymptotically faster than real wages, even though both grow faster than labor-augmenting technological progress:

$$\lim_{t \rightarrow \infty} g_{Y/N} > \lim_{t \rightarrow \infty} g_w > \gamma.$$

Labor's income share therefore converges to zero, while the AI industry's revenue as a share of output converges to $1 - \alpha$.

Figure 3 shows where the long-run regimes characterized in Proposition 1 lie relative to $\mathcal{B}(\sigma)$. For $0 < \sigma < 1$, the equilibria above the threshold have output-per-worker growth converging to γ and a positive limiting labor income share. For $\sigma > 1$, these same long-run outcomes occur below the threshold. Above it, the regime is AI-dominated: output per worker grows asymptotically faster than γ and labor's income share converges to zero.

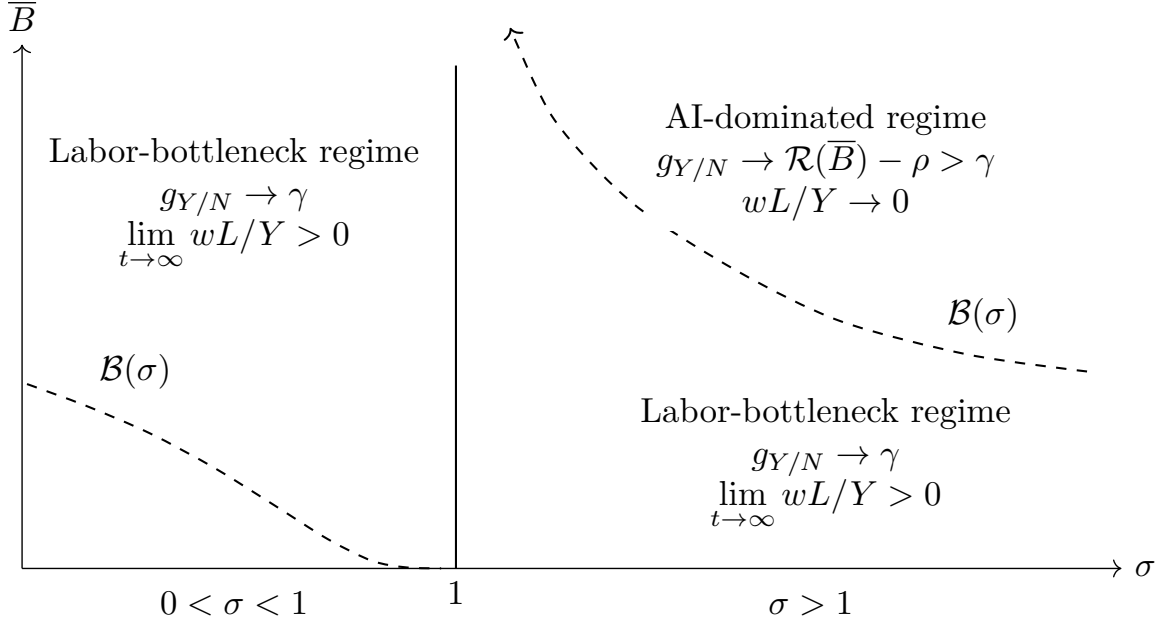


Figure 3: Constructed regimes with a finite upper bound. The vertical axis is linear and the curves are schematic rather than calibrated. For $0 < \sigma < 1$, $\bar{B} > \mathcal{B}(\sigma)$ is the condition for the positive-labor-share construction in Proposition 1(i). For $\sigma > 1$, the curve separates the labor-bottleneck and AI-dominated equilibria in Proposition 1(iii) and 1(iv).

Table 1 summarizes the four cases of Proposition 1. For $\sigma > 1$, an upper bound below $\mathcal{B}(\sigma)$ makes the return on capital at a vanishing labor share too low for capital and AI services to outgrow effective labor. Above the threshold, that return supports their faster growth along the AI-dominated equilibria. Across these equilibrium regimes, long-run real-wage growth exceeds γ if and only if labor's income share converges to zero.

Table 1: Long-run equilibrium characterization

Elasticity	AI efficiency bound	$\lim g_{Y/N}$	$\lim g_w$	$\lim wL/Y$	$\lim r$
$0 < \sigma < 1$	$\bar{B} > \mathcal{B}(\sigma)$	γ	γ	Positive	$\rho + \gamma$
$\sigma = 1$	Any finite $\bar{B}$	γ	γ	$(1 - \alpha)\omega_L$	$\rho + \gamma$
$\sigma > 1$	$\bar{B} < \mathcal{B}(\sigma)$	γ	γ	Positive	$\rho + \gamma$
$\sigma > 1$	$\bar{B} > \mathcal{B}(\sigma)$	$\mathcal{R}(\bar{B}) - \rho$	$\gamma + \frac{\mathcal{R}(\bar{B}) - \rho - \gamma}{\sigma}$	Zero	$\mathcal{R}(\bar{B})$

5 Equilibrium without an upper bound on AI efficiency

I now remove the upper bound on AI efficiency and compare the three substitution regimes. The RSI technology becomes

$$\dot{B} = \chi(BM)^\eta. \quad (42)$$

All other primitives remain as in Section 3. In Definition 1, I set $\psi(B) = 1$ and $\psi'(B) = 0$ and remove the upper restriction on B . Equilibrium still requires finite allocations at every finite date,

global agent optimality, and both transversality conditions. Removing the upper bound does not by itself imply that B grows without bound.

The results differ in scope across substitution regimes. Under complementarity, output-per-worker growth is bounded by the rate of labor-augmenting technological progress, even though there is no upper bound on AI efficiency. At unit elasticity, I give sufficient parameter restrictions for an analytically tractable balanced-growth equilibrium. Under substitution, I show how growth and interest rates increase with the efficiency upper bound and derive a return bound that applies at every date.

5.1 Complementarity

When $\sigma < 1$, effective labor remains necessary even if AI becomes arbitrarily efficient. Equation (8) gives an upper bound for Z and Y which does not depend on B :

$$Z \leq \omega_L^{\sigma/(\sigma-1)} AL, \quad (43)$$

$$Y \leq \omega_L^{\sigma(1-\alpha)/(\sigma-1)} K^\alpha (AL)^{1-\alpha}. \quad (44)$$

Proposition 2 (Production bounds under complementarity). *Let $0 < \sigma < 1$ and $\bar{B} = \infty$. Along any feasible path satisfying the production equations (7) and (8), labor-market clearing $L = N$ in equation (28a), the uncapped RSI technology (42), and the resource constraint (26), the following holds:*

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log \left(\frac{Y_t/N_t}{Y_0/N_0} \right) \leq \gamma. \quad (45)$$

Proof. See Appendix A, Proof of Proposition 2. □

Thus a permanent growth rate of output per person above γ is infeasible, although transitional growth can exceed it. Complementarity suffices to bound long-run growth despite RSI and no bound on AI efficiency.

Proposition 3 (Long-run outcomes under complementarity without an AI efficiency upper bound). *Suppose $0 < \sigma < 1$, $\bar{B} = \infty$, and an equilibrium satisfies $B \rightarrow \infty$. Then*

$$\frac{K}{Y} \rightarrow \frac{\alpha}{\rho + \gamma + \delta} > 0, \quad \frac{C}{Y} \rightarrow 1 - \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} > 0, \quad (46)$$

$$\frac{U}{Y} \rightarrow 0, \quad \frac{M}{Y} \rightarrow 0, \quad (47)$$

$$g_{Y/N} \rightarrow \gamma, \quad g_w \rightarrow \gamma, \quad r \rightarrow \rho + \gamma, \quad (48)$$

$$s_X \rightarrow \frac{1 - \sigma}{1 - \alpha\sigma}, \quad (49)$$

$$\frac{wL}{Y} \rightarrow \frac{\sigma(1 - \alpha)^2}{1 - \alpha\sigma} > 0, \quad \frac{p_X X}{Y} \rightarrow \frac{(1 - \alpha)(1 - \sigma)}{1 - \alpha\sigma} > 0. \quad (50)$$

Proof. See Appendix A, Proof of Proposition 3. □

Inference and research compute vanish as shares of output, not necessarily in levels. The AI industry nevertheless retains a positive revenue share.

5.2 Unit elasticity

At unit elasticity, production becomes

$$Y = K^\alpha (AL)^{(1-\alpha)\omega_L} X^{(1-\alpha)\omega_X}. \quad (51)$$

Consider a balanced-growth path (BGP) with positive constant K/Y , C/Y , and M/Y . The AI-demand and monopoly conditions, equations (13) and (19), imply $U/Y = (1 - \alpha)^2 \omega_X^2$. Along the BGP, $g_K = g_U = g_M = g_Y$. Substituting $X = BU$ into production, equation (51), and log-differentiating this equation and the RSI technology in equation (42) yields

$$g_Y = n + \gamma + \frac{\omega_X}{\omega_L} g_B, \quad (52)$$

$$(1 - \eta) g_B = \eta g_Y. \quad (53)$$

The first relation describes how AI improvement raises production growth; the second describes how expanding research resources sustain AI improvement. Solving these relations gives candidate growth rates. The next proposition establishes that they belong to an equilibrium.

Proposition 4 (An uncapped balanced-growth equilibrium and local convergence). *Suppose $\sigma = 1$, $\bar{B} = \infty$, and*

$$0 < \eta < \alpha, \quad (1 - \alpha)\omega_X \leq \frac{1}{2}, \quad \eta + \omega_X < 1. \quad (54)$$

(i) *There is an equilibrium trajectory such that*

$$g_{Y/N} = g_w = \gamma + \frac{\eta\omega_X}{1 - \eta - \omega_X} (n + \gamma), \quad (55)$$

$$r = \rho + \gamma + \frac{\eta\omega_X}{1 - \eta - \omega_X} (n + \gamma), \quad (56)$$

$$\frac{wL}{Y} = (1 - \alpha)\omega_L > 0, \quad (57)$$

$$g_B = \frac{\eta(1 - \omega_X)}{1 - \eta - \omega_X} (n + \gamma). \quad (58)$$

(ii) *There is an open neighborhood of the initial condition (A_0, N_0, K_0, B_0) of the BGP in part (i) such that, from every initial condition in that neighborhood, there exists an equilibrium trajectory converging to the corresponding BGP in normalized variables.*

Proof. See Appendix A, Proof of Proposition 4. □

Equations (55) and (58) quantify the two feedback loops. In the technological loop, a higher B makes each unit of research compute more productive through BM . In the economic loop, better

AI raises output, allowing more resources to be devoted to research. Along this BGP, the developer optimally chooses a positive constant M/Y , so research compute grows with output. Despite autonomous research, g_B remains proportional to the growth rate of effective labor, $n + \gamma$. Human labor no longer enters the RSI technology, but at unit elasticity it remains essential to final-good production. Growth in effective labor expands the resources available for research, which raises B and further expands output. The common denominator $1 - \eta - \omega_X$ captures this reinforcement: a higher η strengthens the research feedback, while a higher ω_X makes output growth more responsive to AI improvement. Within the stated restrictions, both raise g_B and the growth premium above γ . Real wages share this premium because unit elasticity keeps labor's income share constant.

5.3 Substitution: growth and the AI-efficiency upper bound

With $\sigma > 1$, Proposition 1 establishes AI-dominated equilibria whose limiting interest rate and output-per-worker growth rate are $\mathcal{R}(\bar{B})$ and $\mathcal{R}(\bar{B}) - \rho$. For each fixed $\sigma > 1$, both increase without bound as $\bar{B} \rightarrow \infty$. This contrasts with Proposition 2: under complementarity, the long-run growth bound does not depend on an AI-efficiency upper bound.

The same function also bounds the interest rate at every date, with or without an efficiency upper bound:

$$r_t \geq \mathcal{R}(B_t), \quad \sigma > 1. \quad (59)$$

To see this, combine equations (32), (33), and (34) before taking the limit $s_X \rightarrow 1$:

$$\frac{Y}{K} = \left[B(1 - \alpha)\omega_X^{\sigma/(\sigma-1)}(1 - e_X)s_X^{-1/(\sigma-1)} \right]^{(1-\alpha)/\alpha}. \quad (60)$$

For fixed B , this expression decreases with s_X and reaches its lower limit as $s_X \rightarrow 1$.³ Substitution into $r = \alpha Y/K - \delta$ gives equation (59). Thus $\mathcal{R}(B)$ is the lowest return compatible with the production and monopoly conditions at that efficiency level, attained in the zero-labor-share limit.

If B grows without bound along a trajectory, this inequality requires $r \rightarrow \infty$ and $Y/K \rightarrow \infty$. Production also gives $Z/Y = (Y/K)^{\alpha/(1-\alpha)} \rightarrow \infty$, while the Euler equation (5) requires $g_{C/N} = r - \rho \rightarrow \infty$. These conditional implications and the comparison across finite upper bounds do not establish a finite-time singularity or settle equilibrium existence without an upper bound.

The developer's problem distinguishes the cases $\eta < \alpha$ and $\eta > \alpha$.

³Using $e_X = (1 - s_X)/\sigma + \alpha s_X$ from equation (15),

$$\frac{d \log[(1 - e_X)s_X^{-1/(\sigma-1)}]}{d \log s_X} = \frac{(\sigma - 2)(1 - \alpha\sigma)s_X - (\sigma - 1)}{\sigma(\sigma - 1)(1 - e_X)} < 0.$$

The denominator is positive for $\sigma > 1$. The numerator is affine in s_X and negative at both endpoints: $-(\sigma - 1)$ at zero and $-[1 - \alpha + \alpha(\sigma - 1)^2]$ at one.

Proposition 5 (Research expenditure and the developer’s value). *Suppose $\sigma > 1$, $\bar{B} = \infty$, $0 < \alpha < 1$, and $0 < \eta < 1$. Fix any $B_0 > 0$, any finite horizon $T > 0$, positive continuous paths of K, A, L , and an integrable interest-rate path. For the developer’s problem (21) restricted to $[0, T]$:*

- (i) *If $\eta < \alpha$, discounted net profit is bounded above over all research plans and tends to minus infinity as total research expenditure over $[0, T]$ tends to infinity.*
- (ii) *If $\eta > \alpha$, the developer can make discounted net profit arbitrarily large. Consequently, there is no infinite-horizon equilibrium with a finite-valued developer optimum.*

Proof. See Appendix A, Proof of Proposition 5. □

When $\eta < \alpha$, research costs eventually dominate profit gains on a fixed horizon. When $\eta > \alpha$, expanding early research makes the gains outgrow the cost. Each deviating plan uses finite research expenditure and keeps AI efficiency finite on every finite interval. The obstruction is therefore an unbounded objective, not a finite-time singularity.

The first case motivates the choice $\eta = 0.20 < \alpha = 0.33$ in Section 6: it avoids this unbounded-payoff mechanism even without an efficiency cap. This is a modeling restriction, not an estimate of η . With a finite upper bound, $\eta < \alpha$ is not needed to bound the developer’s finite-horizon value. Part (i) does not establish an infinite-horizon equilibrium without an upper bound. At $\eta = \alpha$, comparing the growth orders of benefits and costs alone is inconclusive.

6 Quantitative equilibrium paths

6.1 Experimental design and initial conditions

I study an unanticipated activation of RSI, rather than the first introduction of AI. Before date zero, the economy already uses AI production services, with $\omega_X = 0.10$ and fixed efficiency B_0 , but research that improves AI efficiency is unavailable. It follows the BGP of this fixed-efficiency economy. At date zero, the developer gains access to the RSI technology with $\chi > 0$. Neither the production function nor its weights change. The pre-event restriction corresponds to $\chi = 0$ and $M = 0$; it is not an imposed zero-research choice under the post-event technology.

I compare $\sigma = 0.90, 1, 1.10, 1.50$, retaining $\eta = 0.20$, $B_0/\bar{B} = 0.01$, and $\bar{B} = 1.10\mathcal{B}(1.50)$, with the threshold evaluated at $\omega_X = 0.10$. The upper bound is chosen to put $\sigma = 1.50$ in the AI-dominated regime; it is not estimated. The other three elasticities converge to labor-supported regimes. The choice $\eta = 0.20 < \alpha = 0.33$ is motivated by Proposition 5(i): on a fixed horizon, research costs eventually dominate profit gains even without an upper bound on AI efficiency. It is an illustrative restriction, not an empirical estimate or a necessary condition for bounded value with the finite upper bound used here. The following subsections differ only in the calibration of χ : research expenditure guides the principal exercise, and a price-decline target gives a faster-transition sensitivity. Each subsection reports the full parameter and initial-stock table, including sources and assumed values.

For each elasticity, I choose the pre-event capital stock to satisfy

$$r_0 = \rho + \gamma, \quad \frac{K_0}{Y_0} = \frac{\alpha}{\rho + \gamma + \delta} = 3.30. \quad (61)$$

Here Y_0 is obtained from the production and monopoly conditions with AI already present. Thus K_0 differs across elasticities, while the initial capital–output ratio and interest rate are common. With $A_0 = N_0 = 1$ and the common B_0 , the four capital stocks are approximately 3.434, 4.649, 5.037, and 5.505. These are calibrated alternative economies, not different outcomes of introducing AI into one economy with the same capital stock. The ratio is a theoretical BGP reference, not an exact empirical fit: the PWT 11.0 comparison yields a 2023 current-price capital–output ratio of about 3.61 for the covered world sample and 3.42 for the United States, with capital definitions that differ from the one-good model (Feenstra et al., 2025).

Before activation, consumption satisfies

$$C(0^-) = Y_0 - U_0 - (\delta + n + \gamma)K_0 > 0. \quad (62)$$

Only K_0 and B_0 , together with the exogenous A_0, N_0 , are inherited by the post-event equilibrium. The BVP selects $C(0^+)$ and $q(0^+)$ anew. Since the production block and its predetermined inputs do not change, Y, X, U, w, r , and p_X are continuous at activation. Research, consumption, investment, and growth rates can adjust immediately. The pre-event BGP requires an unanticipated activation: announcing RSI earlier would generally change saving and consumption before date zero, as in the technology-news mechanisms reviewed by Beaudry and Portier (2014).

The sudden activation is an experimental assumption, not a requirement of the RSI technology. It differs from the continued endogenous automation in Jones and Tonetti (2026). The exercise also omits the costs of expanding compute capacity in GATE (Erdil et al., 2025) and the complementary intangible investment studied by Brynjolfsson et al. (2021). Those mechanisms can delay adjustment or measured productivity gains, but they do not explain the paths computed here. Reducing χ does not remove the unanticipated activation or require consumption and research spending to be continuous at date zero. Appendix B describes the equilibrium solver, numerical checks and replication files.

6.2 Principal calibration: research expenditure

I use research expenditure to guide the choice of χ . The reference is the United States in 2023: Korinek and McKelvey (2026) estimate training compute expenditure of \$18.46 billion, equivalent to 0.0664% of nominal GDP.⁴ This is the lowest of the 2023–2025 estimates considered, not a lower confidence bound or an estimate of current research spending. The compute estimate uses rental-equivalent costs and assigns half of AI compute spending to training and research. It proxies M but does not measure autonomous RSI directly.

⁴Training compute is from Table 3 of Korinek and McKelvey (2026). Nominal GDP of \$27,811.517 billion is the 2023 value of the BEA series GDPA, available through FRED, accessed September 14, 2026.

This exercise compares $\sigma = 0.90, 1, 1.10, 1.50$, starting each economy on its fixed-efficiency pre-RSI BGP from Section 6.1. AI is already used in production, $B_0 = 0.01\bar{B}$, and initial capital satisfies $K_0/Y_0 = 3.30$. RSI becomes available unexpectedly at date zero, without changing the initial stocks, production quantities or prices. Consumption and research adjust endogenously. The initial capital stocks differ across elasticities to preserve the common capital–output ratio, not to fit post-activation outcomes. All parameters except χ are shared with the sensitivity exercise in Section 6.3; Table 2 lists them.

At $\sigma = 1$, I compare annual expenditure with annual output:

$$\frac{\int_0^1 M_t dt}{\int_0^1 Y_t dt} \simeq \frac{18.46}{27,811.517} = 0.00066375. \quad (63)$$

Among the previously explored candidates on the increasing low- χ branch, $\chi = 1.4378$ gives the closest annual ratio, 0.06374%. It falls short of the empirical proxy by 0.26 basis points, or about 4.0% of the target. I retain this approximate match rather than refine χ to fit an uncertain proxy more closely. This choice does not relax the equilibrium checks in Appendix B. I then hold $\chi = 1.4378$ fixed across the other three elasticities. Their research expenditure and price declines are equilibrium outcomes, not separately fitted targets.

The initial growth effects are modest across all four elasticities. Output per person begins growing at instantaneous annual rates of 1.08%, 1.23%, 1.26%, and 1.45%, compared with 1% before activation. At $\sigma = 1.50$, consumption rises by 0.84%, research initially absorbs 0.57% of output, and developer net profit is positive at 2.73% of output. Gross investment remains 19.57% of output: physical capital grows by 0.93% annually at activation, although it falls relative to effective labor, which grows by 1.30%. This adjustment requires neither initial developer losses nor capital liquidation.

The first-year research shares are 0.00687%, 0.06374%, 0.14175%, and 0.56313% of output. Only the second value is calibrated: stronger substitution changes the endogenous incentive to invest in research. The untargeted price declines over 27 months are approximately 2.4%, 9.0%, 10.9%, and 14.2%. Research expenditure relative to output falls between the first and second years in all four cases, so matching its level at unit elasticity does not reproduce its recent empirical growth.

Figures 4–6 show growth rates, prices and income shares, using separate windows for the first ten years and years 10–500. The subsequent paths separate: the three labor-supported economies return to output-per-person and real-wage growth of 1% and an interest rate of 5%. At $\sigma = 1.50$, growth accelerates and labor’s income share tends to zero. Its decline reaches half its initial share after approximately 221 model years. The limiting output-per-person growth, wage growth and interest rate are 3.13%, 2.42%, and 7.13%, respectively.

The plotted horizon is not the terminal boundary. At year 500 in the $\sigma = 1.50$ case, output-per-person growth is 4.27%, wage growth is 3.19% and the interest rate is 8.20%, still above their limits; labor’s income share is 2.64%. The numerical solution is extended far beyond the displayed window to verify convergence. At unit elasticity, $B/\bar{B}$ is only 0.46 at year 500, while labor’s income

share remains 60.3%. A slow approach to the efficiency bound need not imply a large remaining deviation of growth from its limiting rate.

Table 2: Principal calibration: parameters and initial stocks

Symbol	Value	Interpretation	Source or justification
α	0.33	Output elasticity with respect to capital	Assumed one-third macro benchmark; not estimated here.
δ	0.05	Capital depreciation rate	Assumed annual benchmark; PWT provides a comparison, not a fitted target (Feenstra et al., 2025).
ρ	0.04	Household discount rate	Assumed preference parameter; not estimated.
n	0.003	Population growth rate	Rounded constant-rate approximation to the UN 2024–2100 world projection, not a perpetual UN forecast (United Nations, 2024).
γ	0.01	Growth of labor-augmenting technology	Arbitrary trend assumption. OECD labor-productivity projections guide its magnitude but do not measure A (OECD, 2025).
η	0.20	Research exponent	Illustrative, not estimated. $\eta < \alpha$ is motivated by Proposition 5(i); also satisfies $\eta \leq 1/2$.
ω_X	0.10	AI production weight	Assumed value; not fitted to industry revenue.
ω_L	0.90	Effective-labor weight	Computed as $1 - \omega_X$.
χ	1.4378	Research productivity after activation	Approximate expenditure calibration at $\sigma = 1$: 0.06374% against 0.06638% (2023 US proxy); fixed across elasticities.
σ	0.90; 1; 1.10; 1.50	AI–labor substitution	Arbitrary scenarios, not estimates.
$\bar{B}$	1,360.992139	AI-efficiency upper bound	1.10 $\bar{B}$ (1.50); arbitrary scenario, not estimated.
A_0, N_0	1; 1	Initial technology and population	Same unit normalizations.
B_0	13.609921	Initial AI efficiency	0.01 $\bar{B}$; same arbitrary initial distance.
K_0	3.433645; 4.649320; 5.036956; 5.504667	Initial capital, by σ	Same fixed-efficiency BGP stocks with existing AI; Equation (61).
K_0/Y_0	3.30	Initial capital–output ratio	Derived from the pre-RSI BGP and preserved at activation.

Notes: Rates are annual. Values are rounded for display; the simulations retain their original precision. $L_0 = N_0$. Initial consumption C_0 and the developer’s shadow value q_0 are selected endogenously by the BVP for each path, not supplied as inputs. K_0/Y_0 is a derived ratio, measured in years of date-zero output, not an additional parameter. Lists of four ratios follow the displayed order of σ .

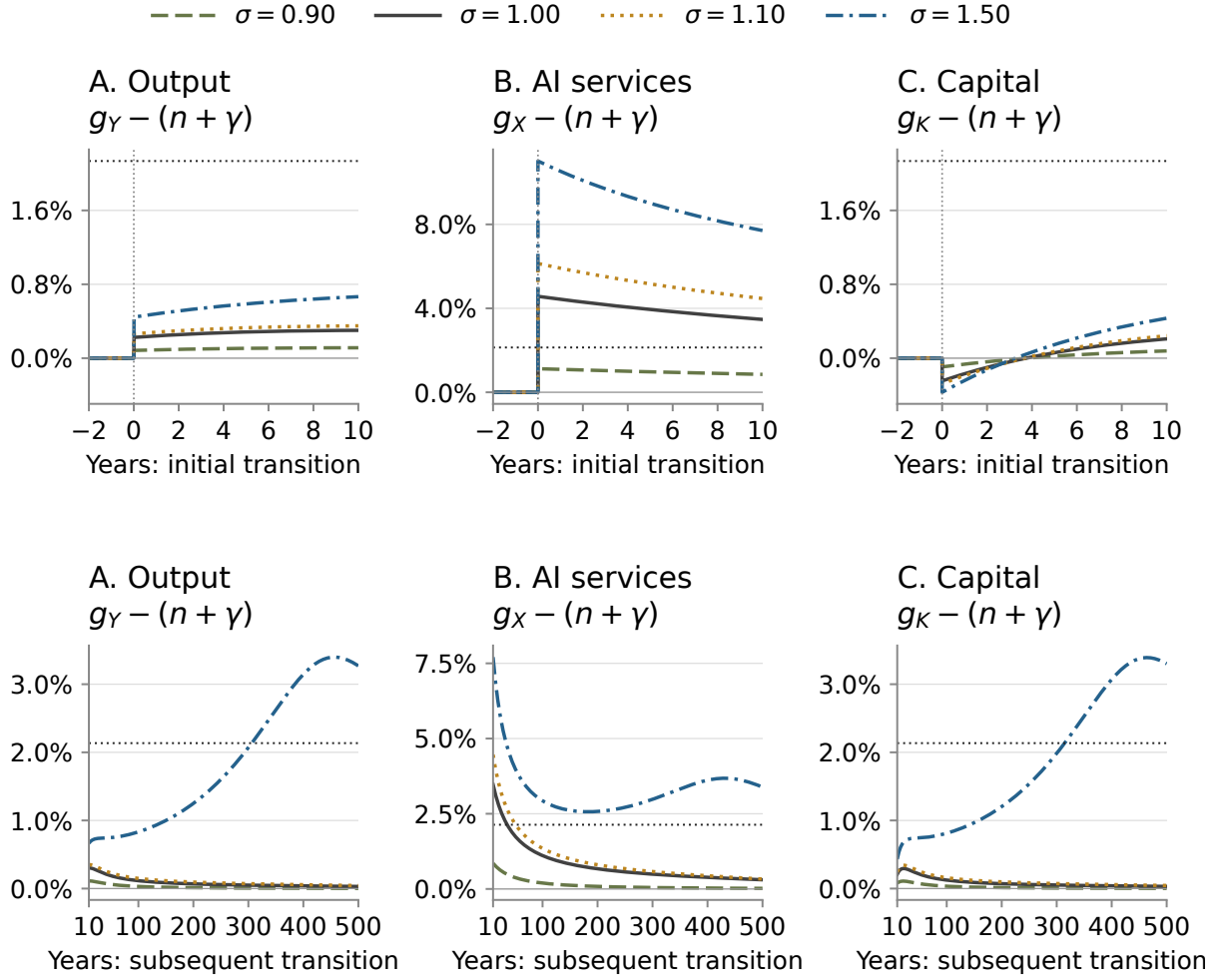


Figure 4: Principal calibration: growth of output, AI production services and capital per unit of effective labor. Rates are instantaneous annual rates. The upper row shows years -2 to 10 and the lower row years 10 – 500 . Vertical scales differ between rows; output and capital share a scale within each row. The vertical line marks RSI activation, and dotted horizontal lines mark the $\sigma = 1.50$ limits.

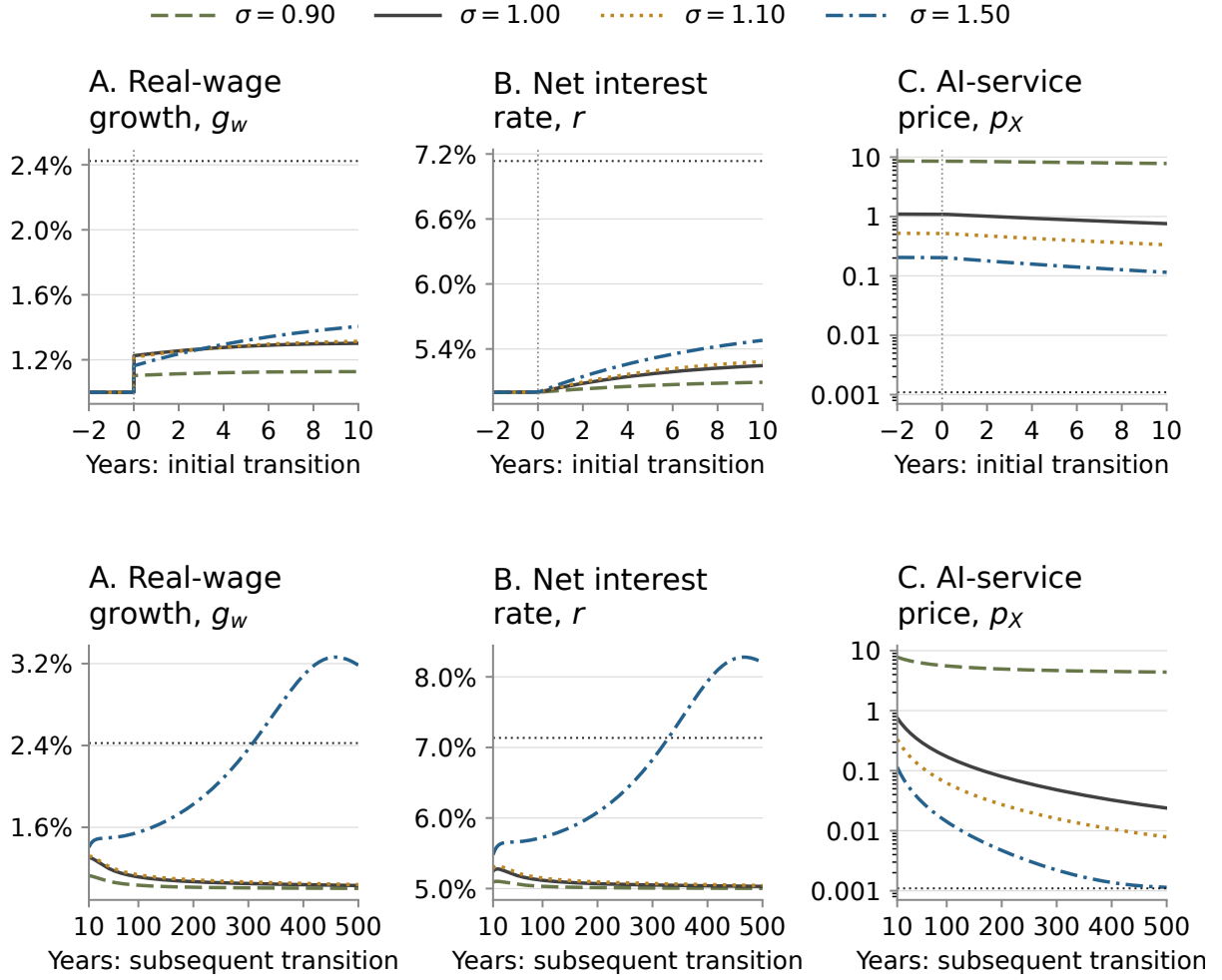


Figure 5: Principal calibration: real-wage growth, the net interest rate and the AI-service price. The price is in final-good units on a logarithmic scale and is not a calibration target. Vertical scales differ between the initial and subsequent windows. Dotted horizontal lines mark the $\sigma = 1.50$ limits.

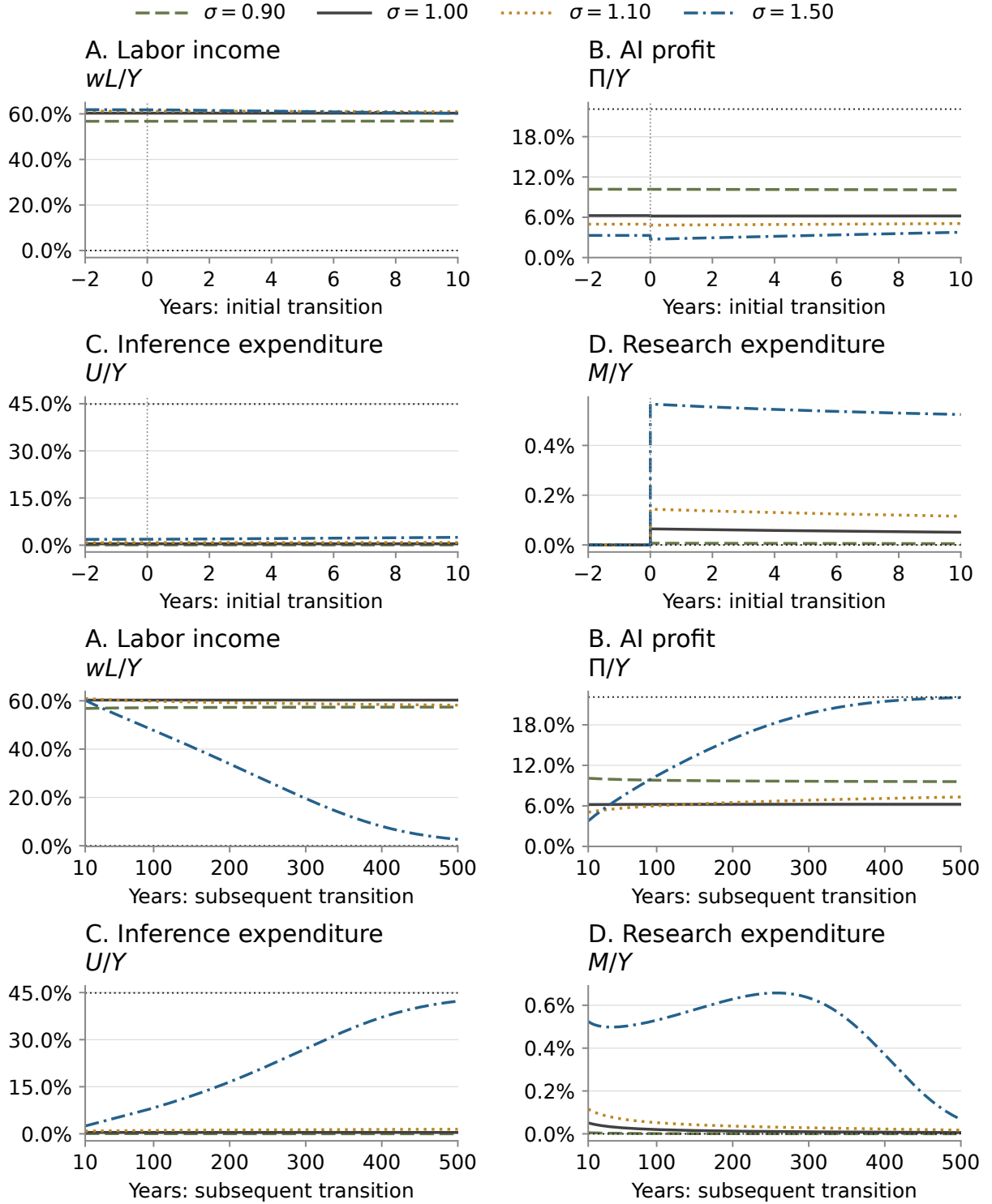


Figure 6: Principal calibration: labor income, AI net profit, inference expenditure and research expenditure as shares of output. The first two rows show years -2 to 10 and the last two show years 10–500. The instantaneous research share differs from the annual ratio in Equation (63). Dotted horizontal lines mark the $\sigma = 1.50$ limits; the gross capital share remains 0.33.

6.3 Sensitivity: a faster transition

I next calibrate χ to half the nearly 80% decline in the OECD quality-adjusted language-model price index between January 2024 and April 2026 (OECD, 2026). At $\sigma = 1$, the target is

$$\frac{p_X(2.25)}{p_X(0)} = 1 - \frac{0.80}{2} = 0.60. \quad (64)$$

The target is a 40% decline over 2.25 years. It halves the percentage decline, not its logarithm, the annual growth rate, or χ itself. This is an illustrative sensitivity exercise, not a second empirical estimate or an estimate of how monopoly changes the pace of price declines. Competition, hardware, human research and changes in task intensity also affect observed prices. The comparison treats the final-good price as constant over this window. Date zero remains a hypothetical RSI activation, not a claim that it occurred in January 2024.

At unit elasticity, Equation (19) gives a constant markup and a price inversely proportional to B . The target therefore requires B to increase by a factor of 5/3 within 2.25 years.

All other parameters and initial stocks are unchanged. Each economy starts on the same fixed-efficiency BGP as in Section 6.1, with existing AI and $K_0/Y_0 = 3.30$. RSI is activated unexpectedly at date zero. I recalibrate χ at unit elasticity, hold it fixed across the other three elasticities, and solve consumption and research again using the same BVP and equilibrium checks. The new paths are not time-rescaled versions of the earlier paths. Table 3 reports the parameters and initial stocks.

Table 3: Faster-transition sensitivity: parameters and initial stocks

Symbol	Value	Interpretation	Source or justification
α	0.33	Output elasticity with respect to capital	Assumed one-third macro benchmark; not estimated here.
δ	0.05	Capital depreciation rate	Assumed annual benchmark; PWT provides a comparison, not a fitted target (Feenstra et al., 2025).
ρ	0.04	Household discount rate	Assumed preference parameter; not estimated.
n	0.003	Population growth rate	Rounded constant-rate approximation to the UN 2024–2100 world projection, not a perpetual UN forecast (United Nations, 2024).
γ	0.01	Growth of labor-augmenting technology	Arbitrary trend assumption. OECD labor-productivity projections guide its magnitude but do not measure A (OECD, 2025).
η	0.20	Research exponent	Illustrative, not estimated. $\eta < \alpha$ is motivated by Proposition 5(i); also satisfies $\eta \leq 1/2$.
ω_X	0.10	AI production weight	Same assumed value as in Table 2; not fitted to industry size.
ω_L	0.90	Effective-labor weight	Computed as $1 - \omega_X$.
χ	7.616305	Research productivity after activation	Fitted at $\sigma = 1$ to Equation (64); common across elasticities.
σ	0.90; 1; 1.10; 1.50	AI-labor substitution	Same arbitrary scenarios.
$\bar{B}$	1,360.992139	AI-efficiency upper bound	$1.10\bar{B}(1.50)$; unchanged arbitrary margin above the threshold.
A_0, N_0	1; 1	Initial technology and population	Same unit normalizations.
B_0	13.609921	Initial AI efficiency	$0.01\bar{B}$; same arbitrary initial distance.
K_0	3.433645; 4.649320; 5.036956; 5.504667	Initial capital, by σ	Same fixed-efficiency BGP stocks with existing AI; Equation (61).
K_0/Y_0	3.30	Initial capital-output ratio	Same pre-RSI BGP ratio, preserved at activation.

Notes: Rates are annual. Values are rounded for display; the simulations retain their original precision. $L_0 = N_0$. Initial consumption C_0 and the developer’s shadow value q_0 are selected endogenously by the BVP for each path, not supplied as inputs. K_0/Y_0 is a derived ratio, measured in years of date-zero output, not an additional parameter. Lists of four ratios follow the displayed order of σ .

The fitted value is $\chi = 7.616305$, about 5.3 times the principal value of 1.4378. Initial output-per-person growth is 1.62%, 2.71%, 3.02%, and 4.54% across the four elasticities. At $\sigma = 1.50$, it rises from 1.45% in the principal calibration to 4.54%. Output, wages and prices remain continuous at activation, but growth rates adjust immediately. The faster transition therefore retains an abrupt response to the unexpected availability of RSI.

The initial resource reallocation is larger. At $\sigma = 1.50$, consumption rises by 5.05% at activation instead of 0.84%, and research absorbs 3.10% of output instead of 0.57%. Initial developer net profit

falls from 2.73% to 0.20% of output but remains positive. Gross investment is 13.79% of output, compared with 19.57% in the principal calibration. Neither calibration requires initial developer losses or negative gross investment in this case. Investment in the faster transition is nevertheless below depreciation, so the capital stock initially falls at an annualized rate of 0.82%. These are equilibrium outcomes, not added financing or investment constraints.

At $\sigma = 1.50$, labor's income share reaches half its initial value after approximately 47 years rather than 221. It completes 90% of its decline toward zero after about 137 years rather than 424. The limiting growth rates, interest rate and income shares are unchanged. Research productivity changes the speed and shape of adjustment, not these long-run outcomes. The initial size of the AI industry is also unchanged and remains an unfitted feature of the calibration.

Figures 7–9 retain the variables, normalizations, line styles, and time windows of the preceding exercise. Vertical scales may differ between exercises to keep the individual paths legible. Dotted horizontal lines show the unchanged analytical limits for $\sigma = 1.50$.

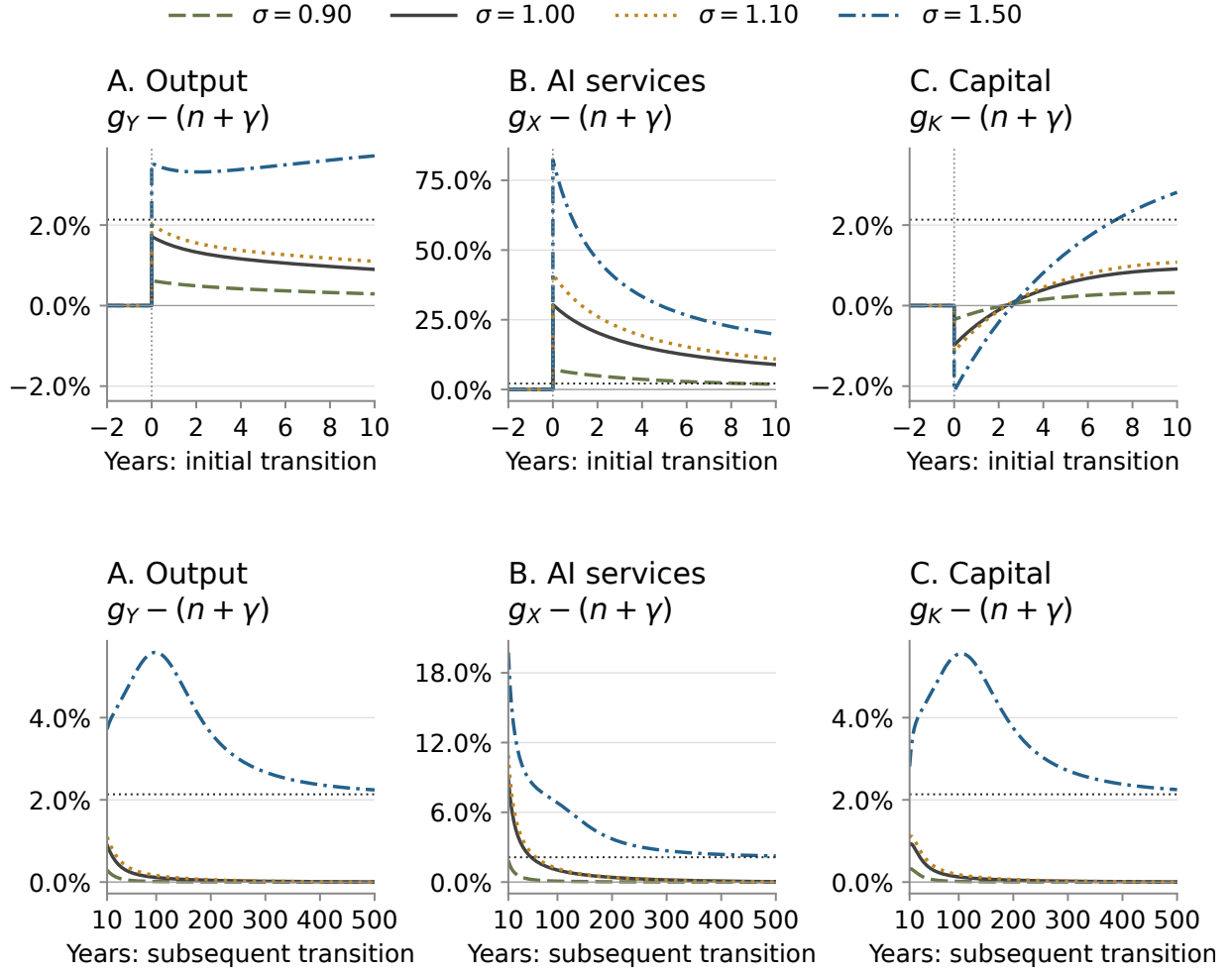


Figure 7: A 40% price-decline target: growth of output, AI production services, and capital per unit of effective labor. Rates are instantaneous annual rates. The upper row shows two pre-event years and years 0–10; the lower row shows years 10–500. Panels A and C share a scale within each row. The vertical line marks RSI activation.

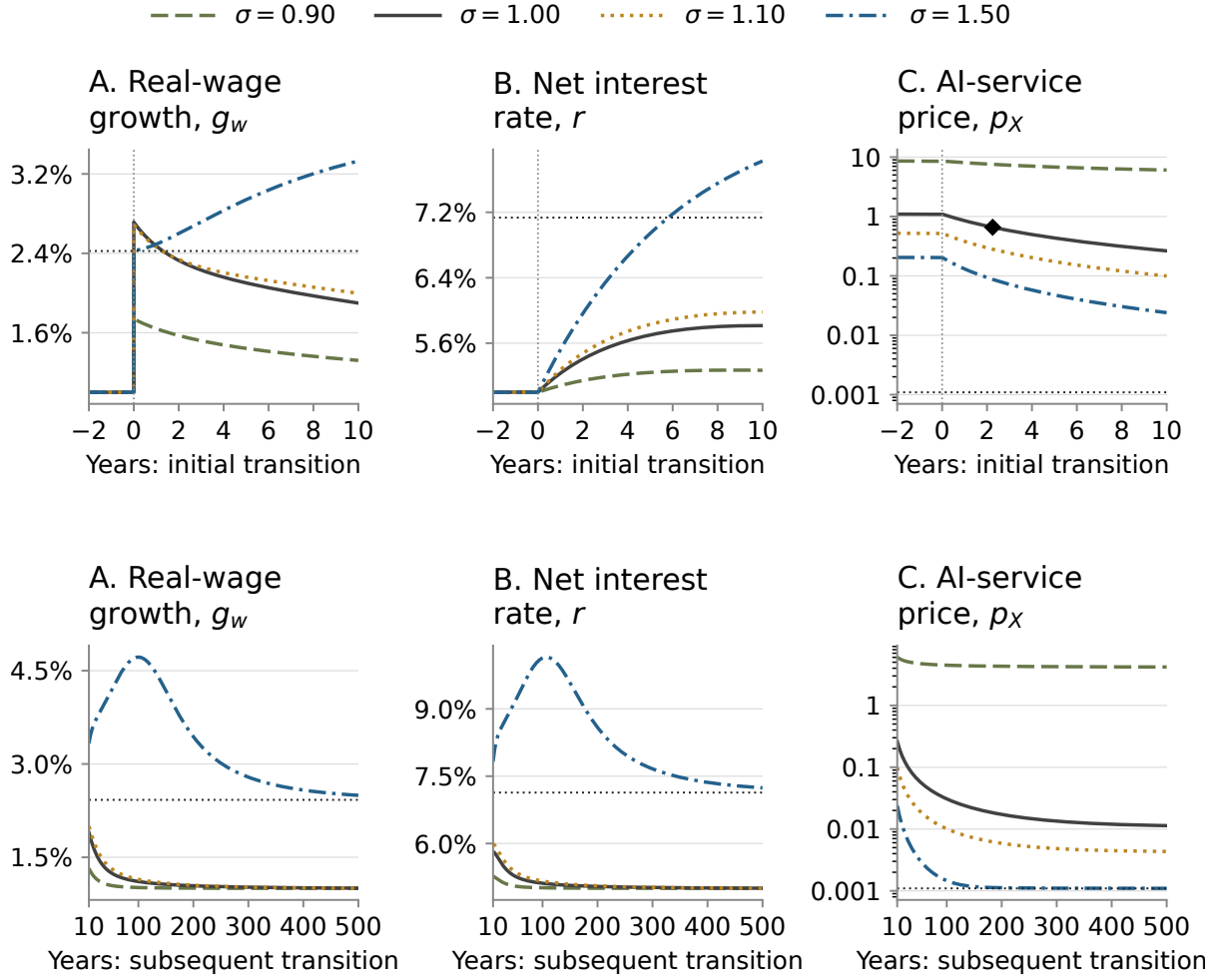


Figure 8: A 40% price-decline target: real-wage growth, the net interest rate, and the AI-service price. Rates are annual; the price is in final-good units on a logarithmic scale. The diamond marks the unit-elastic target $p_X(2.25)/p_X(0) = 0.60$. The vertical line marks RSI activation.

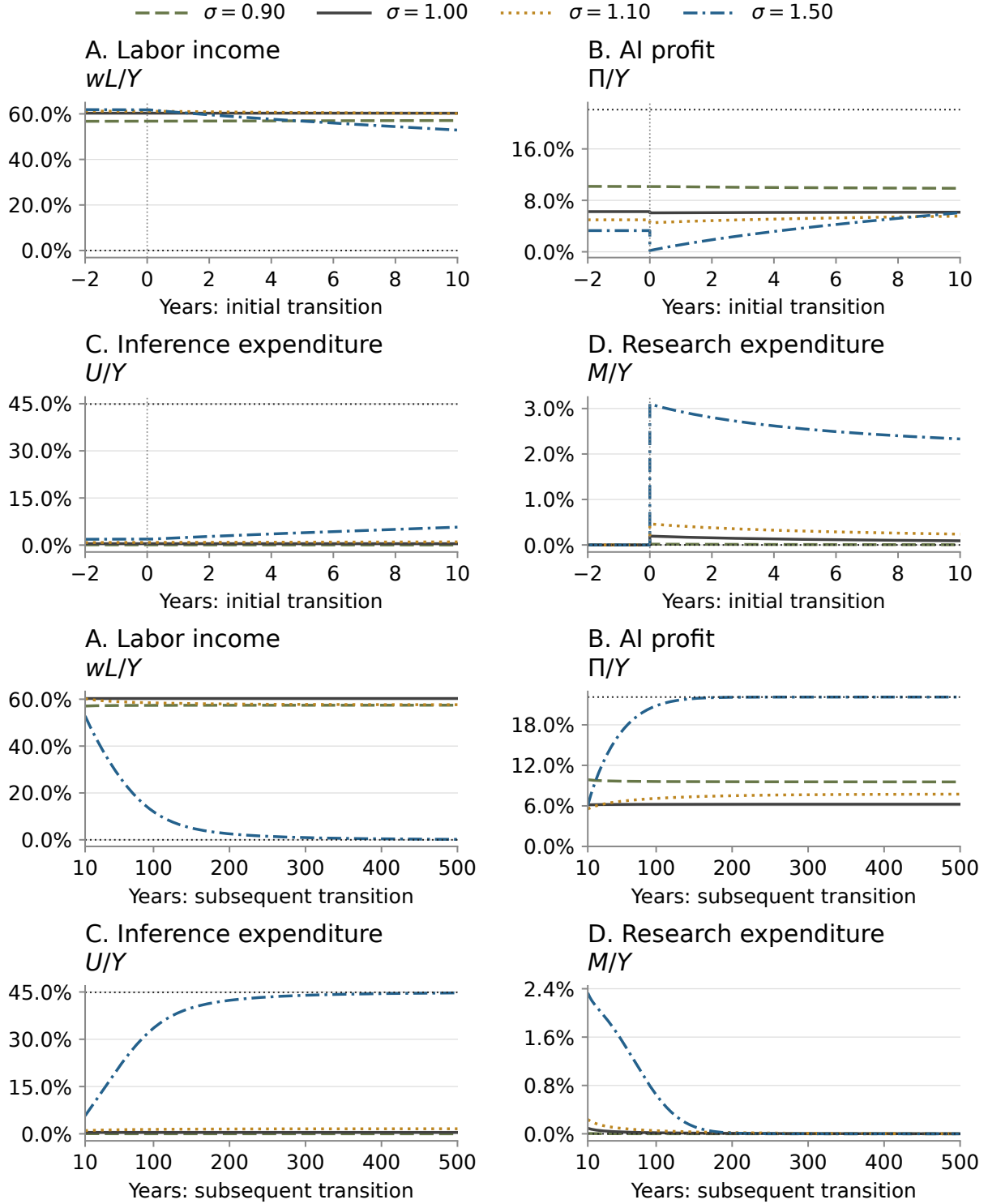


Figure 9: A 40% price-decline target: labor income, AI net profit, inference expenditure, and research expenditure relative to output. The first two rows show the pre-event BGP and years 0–10; the last two show years 10–500. The omitted gross capital share remains $\alpha = 0.33$.

6.4 Interpretation and limitations

The model supports firmer conditional conclusions about long-run growth and income distribution than about takeoff dates. In the finite-upper-bound regimes characterized in Proposition 1, once AI efficiency approaches $\bar{B}$, the limiting growth rates, interest rate and income shares do not depend on $\chi > 0$. They depend on substitutability, the efficiency bound and the other production and macroeconomic parameters. Research productivity instead affects how quickly the economy approaches those limits and how resources are allocated along the way. Its uncertain value makes the reported transition dates particularly uncertain. This distinction does not imply that σ or $\bar{B}$ is well identified: both describe technological scenarios, not estimated probabilities.

The two calibrations are informed by data, not joint empirical fits of the current AI economy. At unit elasticity, AI-industry revenue is $(1 - \alpha)\omega_X = 6.7\%$ of output, independently of K_0 , B_0 and χ . That share is not fitted to the size estimates in [Korinek and McKelvey \(2026\)](#). The expenditure proxy measures historical US training compute, not worldwide autonomous RSI. Matching its level also does not reproduce its recent growth. Nor do the exercises match expenditure and price changes simultaneously. The half-price-decline target is an illustrative sensitivity, not an estimate of the contribution of RSI to observed price changes.

A more aggressive calibration, retained in the replication files, assigns the full rounded 80% observed price decline to RSI and gives $\chi = 36.830702$. At $\sigma = 1.50$, it implies initial output-per-person growth of 23.97%, research spending of 9.32% of output, developer net profit of -6.02% and gross investment of -6.09% of output. Although it passes the numerical equilibrium checks, it requires large developer losses and immediate liquidation of capital. I do not use it as a main quantitative scenario because these initial reallocations are implausibly large. The example exposes the model’s adjustment assumptions: RSI appears unexpectedly, compute expands without installation delays, capital is reversible, and owners can finance losses against future profit. Lower χ moderates the response but does not add the capacity costs or complementary investments in [Erdil et al. \(2025\)](#) and [Brynjolfsson et al. \(2021\)](#). Appendix B links to the preserved exercise and its replication instructions.

Permanent monopoly and the absence of human research further restrict the interpretation of prices and developer profit. The income shares describe payments to labor, capital and the AI developer. Without heterogeneous asset ownership or the creation of new human tasks, they do not establish how gains are distributed across households or how political power changes.

7 Conclusion

The endogenous growth model showed that the long-run economic role of human labor depends critically on whether AI can replace labor’s contribution to production. It characterizes how this substitution margin shapes economic growth, wages, interest rates, and the functional distribution of income when autonomous AI research generates RSI.

In the equilibria studied here with a finite upper bound, complementarity leaves human labor as a bottleneck. Improvements in AI efficiency raise productive capacity but do not permanently increase the growth of output per person or real wages beyond the exogenous rate of labor-augmenting technological progress. Labor therefore retains a positive share of income. The same broad conclusion holds under unit elasticity: AI can support sustained growth, but human labor remains economically essential and continues to receive a positive share of output.

The outcome changes when AI and labor are sufficiently substitutable. Gross substitution is necessary but not sufficient: labor remains a bottleneck while AI services are not productive enough to replace its contribution. Once AI becomes sufficiently efficient, however, capital and effective AI production services can expand faster than effective labor. Output per person and real wages then grow faster than in the labor-bottleneck regime, and the interest rate is higher.

Across the long-run regimes I characterize with a finite upper bound on AI efficiency, real wages grow faster than the exogenous rate only when labor’s income share vanishes. The declining share does not generate faster wage growth. In the AI-dominated regime, output per person grows rapidly enough to raise real wages despite the shrinking fraction paid to labor. Because output per person grows even faster than the real wage, labor’s income share converges to zero. The gross capital share is fixed by the production technology, so this transformation reallocates the remainder of output from labor toward the AI industry. AI-industry revenue should not be confused with monopoly profit: it finances inference and research before the residual is distributed to the developer’s owner.

The uncapped unit-elastic benchmark shows why this qualification matters: continued AI-efficiency growth can raise output per person and real wages together faster than the exogenous rate while labor’s income share remains positive. With complementarity, by contrast, production per unit of effective labor remains bounded even without an upper bound on AI efficiency. This feasibility bound does not establish an equilibrium, but it rules out permanently faster growth of output per person than the exogenous rate.

The numerical exercises give firmer conditional conclusions about the long-run growth regime and income distribution than about takeoff dates. With a finite upper bound, positive research productivity changes the transition but not the limiting outcomes characterized here. Those limits depend on the substitutability between AI and labor, attainable AI efficiency and the other production and macroeconomic parameters. Research productivity is poorly measured, so the reported years should not be read as forecasts. The expenditure-based calibration and the faster price-based sensitivity illustrate this distinction: the economy reaches the same long-run regime through very different paths. Initial AI efficiency and the size of the AI sector also remain important calibration questions.

The analysis points to three lines of future work. A task-based extension with heterogeneous workers, new human tasks, and a continuing role for human researchers could combine the displacement and reinstatement mechanisms of [Acemoglu and Restrepo \(2019\)](#), the labor-market transitions studied by [Korinek et al. \(2026\)](#), and the research-automation network of [Davidson et al. \(2026\)](#). An explicit upstream compute sector, alternative market structures, and heterogeneous asset own-

ership could build on [Korinek and McKelvey \(2026\)](#) and [Korinek and Vipra \(2025\)](#) and translate changes in functional income shares into household inequality and economic power. Finally, a more complete existence analysis could extend the uncapped result beyond unit elasticity, or make the upper bound on AI efficiency endogenous, and determine whether the resulting economy admits an equilibrium. This question is related to, but not resolved by, the explosive-growth mechanisms in [Nordhaus \(2021\)](#) and [Davidson et al. \(2026\)](#).

The central question is whether human labor remains sufficiently essential for wages to keep pace with the economy as a whole.

A Proofs

Lemma 1 (Static AI-service choice). *For $K, A, L, B > 0$, $0 < \alpha < 1$, $\sigma > 0$, and strictly positive CES weights that sum to one, problem (18) has a unique, strictly positive global maximizer. It is characterized by condition (19) and depends smoothly on the positive variables K, A, L, B , for each fixed σ .*

Proof of Lemma 1. Fix $K, A, L, B > 0$, let $R(X) = p_X(X)X$, and write

$$u = \frac{1}{\sigma}, \quad s = s_X, \quad e = u(1 - s) + \alpha s.$$

The demand elasticity in (15) and the CES-share derivative imply

$$R'(X) = p_X(1 - e), \quad \frac{ds}{d \ln X} = (1 - u)s(1 - s).$$

Let $m(X) = R'(X)$. Wherever marginal revenue is positive, direct differentiation gives

$$-\frac{Xm'(X)}{p_X} = e(1 - e) + (\alpha - u)(1 - u)s(1 - s) \equiv D(s). \quad (65)$$

If $(\alpha - u)(1 - u) \geq 0$, then $D(s) > 0$ immediately. If this product is negative, the parameter restrictions imply $0 < \alpha < u < 1$, and

$$D(s) = u(1 - u)(1 - s)^2 + [u^2 + 2\alpha - 3\alpha u]s(1 - s) + \alpha(1 - \alpha)s^2.$$

The middle coefficient is positive. For $u \leq 2/3$, this follows from $u^2 + \alpha(2 - 3u) > 0$. For $u > 2/3$, use $\alpha < u$ and $2 - 3u < 0$ to obtain

$$u^2 + \alpha(2 - 3u) > u^2 + u(2 - 3u) = 2u(1 - u) > 0.$$

Thus marginal revenue decreases strictly whenever it is positive. At a solution of $m(X) = 1/B$, its derivative is strictly negative. The implicit-function theorem therefore makes the maximizing service choice smooth in K, A, L, B wherever it exists.

For $\sigma = 1$, define $\beta = (1 - \alpha)\omega_X$. As $X \downarrow 0$, revenue is proportional to X^{1-u} when $\sigma > 1$, to X^β

when $\sigma = 1$, and to $X^{1-\alpha}$ when $\sigma < 1$. Every exponent lies between zero and one, so $m(X) \rightarrow \infty$. As $X \rightarrow \infty$, marginal revenue converges to zero from above if $\sigma \geq 1$; if $\sigma < 1$, it eventually becomes negative. Hence $m(X) = 1/B$ has exactly one solution. Operating profit increases before this solution and decreases after it. The linear cost X/B dominates revenue as $X \rightarrow \infty$, so this solution is the unique global maximum. Equation (65) also proves the strict second-order condition:

$$e_X(1 - e_X) + (\alpha - \sigma^{-1})(1 - \sigma^{-1})s_X(1 - s_X) > 0. \quad (66)$$

□

Interior research. Lemma 1 establishes $X^* > 0$, and the envelope condition (20) gives $\pi_B = X^*/B^2 > 0$. A marginal increase in AI efficiency therefore raises current optimized profit and weakly raises future AI efficiency under a fixed research policy. At a differentiable optimum its current-value shadow price is strictly positive: $q > 0$. The part of the Hamiltonian that depends on M is

$$-M + q\chi B^\eta M^\eta \psi(B).$$

Its right derivative diverges as $M \downarrow 0$ because $q, B, \psi(B) > 0$ and $0 < \eta < 1$. Therefore $M = 0$ cannot maximize the Hamiltonian at a finite date.

Global developer verification. For a finite upper bound, define research depth as the monotone transformation

$$R_B(B) = -\bar{B} \log \left(1 - \frac{B}{\bar{B}} \right), \quad B(R_B) = \bar{B} \left(1 - e^{-R_B/\bar{B}} \right). \quad (67)$$

It maps $B \in [B_0, \bar{B})$ into $R_B \in [R_{B,0}, \infty)$, where $R_{B,t} = R_B(B_t)$ and $R_{B,0} = R_B(B_0)$, and rewrites the RSI technology as

$$\dot{R}_B = \chi[B(R_B)M]^\eta. \quad (68)$$

Write $\pi_t(B)$ for optimized operating profit evaluated at the date- t values of K, A , and L , and define the shadow value of research depth as $q_{R,t} = q_t \psi(B_t)$. Holding this shadow value fixed, let

$$\hat{\mathcal{H}}_t(\tilde{R}_B) \equiv \max_{\tilde{M} \geq 0} \left\{ \pi_t(B(\tilde{R}_B)) - \tilde{M} + q_{R,t} \chi[B(\tilde{R}_B)\tilde{M}]^\eta \right\}.$$

Lemma 2 (Global verification of the developer's research path). *Let $0 < \eta < 1$ and $\chi > 0$. Fix $0 < B_0 < \bar{B} < \infty$ and positive continuous paths for K, A , and L , and take the locally integrable interest-rate path r as given. Consider a feasible candidate $\{B_t, M_t\}_{t \geq 0}$ for problem (21), with continuously differentiable B_t , interior research $M_t > 0$, and a finite objective. Suppose there is a continuously differentiable costate q_t satisfying the research condition (22), the costate equation (23),*

and the transversality condition (25). If, for every $t \geq 0$ and every $\tilde{R}_B \geq R_{B,0}$,

$$\hat{\mathcal{H}}_t(\tilde{R}_B) \leq \hat{\mathcal{H}}_t(R_{B,t}) + \hat{\mathcal{H}}'_t(R_{B,t})(\tilde{R}_B - R_{B,t}), \quad (69)$$

then the candidate is a global maximizer of the reduced developer problem. Admissible alternatives have $M_t \geq 0$, locally finite research expenditure, and the same initial state; the comparison also bounds the upper limit of their truncated discounted payoffs. Together with the unique static service choice characterized in Lemma 1, it is globally optimal for the developer.

Proof of Lemma 2. Fix a date and the candidate shadow price $q_{R,t} = q_t \psi(B_t) > 0$. Maximizing the research-depth Hamiltonian over research expenditure gives

$$\hat{\mathcal{H}}_t(\tilde{R}_B) = \pi_t(B(\tilde{R}_B)) + \frac{1-\eta}{\eta} M_t \left[\frac{B(\tilde{R}_B)}{B_t} \right]^{\eta/(1-\eta)}. \quad (70)$$

Here M_t is the candidate's research expenditure. The formula follows by maximizing $-M + q_{R,t} \chi[B(\tilde{R}_B)M]^\eta$ over $M \geq 0$ and using the candidate's research first-order condition. Its derivative at the candidate state is

$$\hat{\mathcal{H}}'_t(R_{B,t}) = \frac{\psi(B_t)}{B_t} (U_t + M_t). \quad (71)$$

Write $D_t = \exp(-\int_0^t r_s ds)$ and $\mu_t = D_t q_{R,t}$. The transformed costate equation is $\dot{\mu}_t = -D_t \hat{\mathcal{H}}'_t(R_{B,t})$. For any admissible alternative research policy and its state $\tilde{R}_{B,t}$, maximizing over its control first bounds its Hamiltonian by (70). Apply (69), subtract the candidate payoff, and integrate by parts. Equal initial states imply

$$\tilde{J}_T - J_T \leq -\mu_T (\tilde{R}_{B,T} - R_{B,T}) \leq \mu_T R_{B,T}.$$

Because

$$\psi(B_t) R_{B,t} = -(\bar{B} - B_t) \log \left(1 - \frac{B_t}{\bar{B}} \right) \leq B_t,$$

the last expression converges to zero by the developer's transversality condition. Taking the upper limit establishes global optimality. The argument requires a global upper tangent at the candidate state, not concavity at every counterfactual state. $\square$

A concavity test for Lemma 2. Fix a date t and an alternative AI-efficiency level $b \in [B_0, \bar{B}]$. Let $X_t^*(b)$ be the unique maximizer of problem (18), evaluated at K_t, A_t , and L_t , and let $s_{X,t}(b)$ and $e_{X,t}(b)$ be the corresponding values. Implicitly differentiating the monopoly condition (19), using Equations (10) and (15), gives

$$\varepsilon_{B,t}(b) \equiv \frac{\partial \ln X_t^*(b)}{\partial \ln b} = \frac{1 - e_{X,t}(b)}{e_{X,t}(b)[1 - e_{X,t}(b)] + (\alpha - \sigma^{-1})(1 - \sigma^{-1})s_{X,t}(b)[1 - s_{X,t}(b)]}. \quad (72)$$

This endogenous elasticity is not an additional model parameter. In what follows, suppress the date and counterfactual-state arguments and write

$$X^*(B) \equiv X_t^*(b), \quad \varepsilon_B \equiv \varepsilon_{B,t}(b).$$

Because $\pi_B = X^*(B)/B^2$,

$$\pi_{BB} = \frac{X^*}{B^3}(\varepsilon_B - 2). \quad (73)$$

A direct differentiation gives

$$\frac{\partial^2 \pi(B(R_B))}{\partial R_B^2} = \frac{X^*(B)\psi(B)}{B^3} \left[\psi(B)(\varepsilon_B - 2) - \frac{B}{\overline{B}} \right]. \quad (74)$$

The transformation (67) satisfies

$$B'(R_B) = \psi(B) > 0, \quad B''(R_B) = -\frac{\psi(B)}{\overline{B}} < 0.$$

Using $\pi_B = X^*(B)/B^2$ and (73) gives (74). Hence the algebraic condition

$$\psi(b)[\varepsilon_{B,t}(b) - 2] \leq \frac{b}{\overline{B}} \quad (75)$$

for every $t \geq 0$ and $b \in [B_0, \overline{B})$ makes optimized operating profit concave in research depth throughout the reachable state domain.

The research term in (70) is a positive multiple of $B(R_B)^\mu$, where $\mu = \eta/(1 - \eta) > 0$. Its curvature is

$$\frac{d^2 B(R_B)^\mu}{dR_B^2} = \mu B^{\mu-2} \psi(B) \left[(\mu - 1)\psi(B) - \frac{B}{\overline{B}} \right]. \quad (76)$$

Thus (75), together with

$$\frac{B_0}{\overline{B}} \geq \max \left\{ 0, \frac{2\eta - 1}{\eta} \right\}, \quad (77)$$

makes the maximized Hamiltonian concave throughout the reachable state domain and proves (69). If $\eta \leq 1/2$, condition (77) holds automatically. But for every fixed $0 < \eta < 1$ its right-hand side is strictly below one, so it also holds without this restriction when initial efficiency is sufficiently close to the upper bound. These curvature tests are sufficient, not necessary: they can fail even when the Hamiltonian has the required supporting tangent at the candidate state.

At unit elasticity, the concavity test has a simple primitive form. Define $\beta = (1 - \alpha)\omega_X$. The static monopoly solution implies $\pi_t(B) = a_t B^{\beta/(1-\beta)}$ for some $a_t > 0$. Because $B(R_B)$ is increasing and concave, $R_B \mapsto \pi_t(B(R_B))$ is globally concave whenever $\beta \leq 1/2$. In the benchmark calibration in Table 2, $\beta = 0.067$, so this sufficient condition holds with substantial slack. It is not imposed as an additional maintained assumption; outside this range, the support condition can instead be checked directly.

The AI-dominated continuation supplies an analytical long-run check on the support inequality. For $1 < \sigma \leq 1/\alpha$, the service elasticity defined in Equation (72) satisfies $\varepsilon_{B,t}(b) \leq 1/\alpha$ for every t and b . Suppressing these arguments, subtracting $\alpha(1 - e_X)$ from the denominator in that equation gives

$$(\sigma^{-1} - \alpha)(1 - s_X) [(1 - \sigma^{-1})(1 - s_X) + (\sigma^{-1} - \alpha)s_X] \geq 0.$$

Choose $B_* \in (B_0, \bar{B})$ with $B_*/\bar{B} > \max\{0, (1 - 2\alpha)/(1 - \alpha)\}$. Operating profit is then concave in research depth for $B \geq B_*$. If $\eta \leq 1/2$, the research term in (70) is concave there as well. For a candidate with $B_t \geq B_*$, it consequently suffices to check

$$\hat{\mathcal{H}}_t(R_{B,t}) - \hat{\mathcal{H}}'_t(R_{B,t})(R_{B,t} - R_{B,0}) \geq \hat{\mathcal{H}}_t(R_B(B_*)). \quad (78)$$

Above B_* the tangent bound follows from concavity. Below B_* the Hamiltonian is increasing in B , so its value is at most the right-hand side, whereas the tangent is minimized at $R_{B,0}$.

Along an AI-dominated continuation, the difference between the two sides of (78), divided by Y_t , converges to

$$\alpha(1 - \alpha) \left[1 - \left(\frac{B_*}{\bar{B}} \right)^{(1-\alpha)/\alpha} \right] > 0.$$

To obtain this limit, use $M/Y \rightarrow 0$, $\psi(B)R_B \rightarrow 0$, and $\pi(B_*)/Y \rightarrow \alpha(1 - \alpha)(B_*/\bar{B})^{(1-\alpha)/\alpha}$. Thus the global support condition holds sufficiently late on this continuation even when the stronger curvature condition fails at low counterfactual AI-efficiency levels. The finite transition still requires its own support check.

Proof of Proposition 1. The proof has three tasks. First, Lemma 1 gives a unique, smooth, globally optimal within-date service choice at given states. I combine this choice with the other equilibrium equations to identify candidate long-run values and verify their admissibility, including positive consumption. Finding these values does not yet establish the existence of a trajectory approaching them.

Second, I write the exact dynamic equations in coordinates with finite candidate limits. The linearization identifies the stable directions; I also check that they can accommodate every sufficiently nearby initial state. Counting stable directions alone would not establish this property. The stable-manifold and inverse-function theorems then select the initial consumption and shadow price for those states. They establish convergent solutions of the nonlinear equations for all future time, not merely solutions of their linear approximation.

Third, I reconstruct the original variables and verify feasibility, market clearing, and both transversality conditions. Household concavity and Lemma 2 establish optimality. For the developer, I verify the required curvature throughout the domain of alternative research policies reachable from the initial efficiency. The upper bound makes this possible for initial efficiency sufficiently close to it. These final checks turn the constructed solutions of the first-order conditions into equilibria. The result is local in initial states, not in the time horizon.

Step 1. *Feasible efficiency and candidate long-run values.*

Let $D = \bar{B} - B$. The state equation with a finite upper bound gives

$$\frac{\dot{D}}{D} = -\frac{\chi}{\bar{B}}(BM)^\eta.$$

Integration gives

$$\bar{B} - B_t = (\bar{B} - B_0) \exp \left\{ -\frac{\chi}{\bar{B}} \int_0^t (B_s M_s)^\eta ds \right\} > 0. \quad (79)$$

Local integrability makes the exponent finite at every finite date, so $D_t > 0$.

For the candidate limits, impose $B = \bar{B}$ only in the normalized boundary equations. Actual trajectories will satisfy $B < \bar{B}$ at every finite date.

Write $\theta = (1 - \alpha)/\alpha$. For $\sigma \neq 1$, let $\varphi = (\sigma - 1)/\sigma$ and define

$$m(s) = 1 - \frac{1-s}{\sigma} - \alpha s, \quad \mathcal{H}_\sigma(s) = \left[(1 - \alpha)s m(s) \left(\frac{\omega_X}{s} \right)^{1/\varphi} \right]^\theta.$$

Here $m(s) = 1 - e_X$ is the marginal-revenue factor, not research compute. The static monopoly optimum has $m(s_X) > 0$ by Lemma 1. The CES share, monopoly condition, and production function, equations (28d), (28i), and (28c), imply

$$\frac{X}{AN} = \left[\frac{\omega_L s_X}{\omega_X (1 - s_X)} \right]^{1/\varphi}, \quad \frac{U}{Y} = (1 - \alpha)s_X m(s_X), \quad (80)$$

$$\frac{Y}{K} = B^\theta \mathcal{H}_\sigma(s_X). \quad (81)$$

For example, $Z = X(\omega_X/s_X)^{1/\varphi}$ and $X = B(1 - \alpha)s_X m(s_X)Y$ give the last identity directly.

If $0 < \sigma < 1$, positive marginal revenue requires

$$s > s_{\min} \equiv \frac{1 - \sigma}{1 - \alpha\sigma}.$$

On $(s_{\min}, 1)$, both $m(s)$ and $s^{-1/(\sigma-1)}$ increase strictly. Since

$$\mathcal{H}_\sigma(s)^{1/\theta} = (1 - \alpha)\omega_X^{1/\varphi} m(s)s^{-1/(\sigma-1)},$$

the share map increases from zero to $\mathcal{H}_\sigma(1) = [(1 - \alpha)^2 \omega_X^{1/\varphi}]^\theta$. Thus the equation

$$\frac{\rho + \gamma + \delta}{\alpha} = \bar{B}^\theta \mathcal{H}_\sigma(\bar{s}_X) \quad (82)$$

has exactly one interior solution if and only if $\bar{B} > \mathcal{B}(\sigma)$.

If $\sigma > 1$, the share map decreases strictly on $(0, 1)$. To verify the sign, put $a = \sigma^{-1} - \alpha$, so $m(s) = \varphi + as$. Its logarithmic derivative has the sign of $a(\sigma - 2)s - \varphi$. If $a(\sigma - 2) \leq 0$ this is negative. If $\sigma > 2$ and $a > 0$, $a(\sigma - 2) < (\sigma - 2)/\sigma < \varphi$; if $1 < \sigma < 2$ and $a < 0$, $a(\sigma - 2) < (1 - \sigma^{-1})(2 - \sigma) < \varphi$.

The map therefore decreases from infinity to $\mathcal{H}_\sigma(1)$. Equation (82) has exactly one interior solution if and only if $\bar{B} < \mathcal{B}(\sigma)$. Since $\mathcal{R}(B) = \alpha B^\theta \mathcal{H}_\sigma(1) - \delta$, these calculations also derive (35)–(36). The interior-share condition is therefore $\mathcal{R}(\bar{B}) > \rho + \gamma$ for $\sigma < 1$, but $\mathcal{R}(\bar{B}) < \rho + \gamma$ for $\sigma > 1$.

For either interior-share case, let x, y, u, k, c denote X, Y, U, K, C divided by AN , respectively. Construct the normalized terminal values sequentially:

$$\begin{aligned} \bar{x} &= \left[\frac{\omega_L \bar{s}_X}{\omega_X (1 - \bar{s}_X)} \right]^{1/\varphi}, & \bar{u} &= \frac{\bar{x}}{\bar{B}}, \\ \bar{y} &= \frac{\bar{u}}{(1 - \alpha) \bar{s}_X m(\bar{s}_X)}, & \bar{k} &= \frac{\alpha \bar{y}}{\rho + \gamma + \delta}, \\ \bar{c} &= \bar{y} - \bar{u} - (\delta + n + \gamma) \bar{k}, & \bar{r} &= \rho + \gamma. \end{aligned} \quad (83)$$

At $\sigma = 1$, define $\beta = (1 - \alpha)\omega_X$. The exact static solution is

$$y = \left[(\beta^2 B)^\beta k^\alpha \right]^{1/(1-\beta)}, \quad u = \beta^2 y, \quad x = Bu.$$

The equation $y/k = (\rho + \gamma + \delta)/\alpha$ has exactly one positive solution for every finite $\bar{B} > 0$:

$$\bar{k} = \left[(\beta^2 \bar{B})^\beta \left(\frac{\rho + \gamma + \delta}{\alpha} \right)^{\beta-1} \right]^{1/(1-\alpha-\beta)}. \quad (84)$$

Set $\bar{y} = [(\rho + \gamma + \delta)/\alpha] \bar{k}$, $\bar{u} = \beta^2 \bar{y}$, and $\bar{x} = \bar{B} \bar{u}$, with $\bar{c}, \bar{r}$ as in (83).

Consumption positivity does not require an additional restriction. Positive marginal revenue implies $0 < m(\bar{s}_X) < 1$. Since $0 < \bar{s}_X < 1$, in all three cases

$$\frac{\bar{u}}{\bar{y}} = (1 - \alpha) \bar{s}_X m(\bar{s}_X) < 1 - \alpha, \quad \frac{(\delta + n + \gamma) \bar{k}}{\bar{y}} = \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} < \alpha.$$

It follows that $\bar{c}/\bar{y} > 0$.

Step 2. The three positive-labor-share cases.

The cases in Proposition 1(i), 1(ii), and 1(iii) share one normalization. Divide capital, consumption, and the AI-efficiency shadow price by effective labor, and multiply the remaining efficiency gap by effective labor:

$$k = \frac{K}{AN}, \quad c = \frac{C}{AN}, \quad d = (\bar{B} - B)AN, \quad \tilde{q} = \frac{q}{AN}, \quad \tau = (AN)^{-1}. \quad (85)$$

Dividing by AN removes effective-labor growth. Multiplying the shrinking gap by AN allows it to converge to a positive value: $\dot{d}/d = n + \gamma - \dot{B}/(\bar{B} - B)$. The variable d measures the scaled gap, not its rate of closure. The constants $\bar{k}, \bar{d}$ are the limiting values of k, d ; they are determined by the model, not additional parameters. Also $\dot{\tau} = -(n + \gamma)\tau$, so $\tau \rightarrow 0$. These coordinates leave the model unchanged: $K = k/\tau$ and $B = \bar{B} - d\tau$ at every finite date.

Given k, B , solve the unique static monopoly problem for y, x, u , and set $r = \alpha y/k - \delta$. Define $a = \dot{B}/(\bar{B} - B)$, the proportional rate of closure of the AI-efficiency gap. The research condition gives

$$B = \bar{B} - d\tau, \quad M = \left[\tilde{q} d \chi \eta \frac{B^\eta}{\bar{B}} \right]^{1/(1-\eta)}, \quad a = \chi \frac{B^\eta M^\eta}{\bar{B}}.$$

Substituting (28k) and (29) gives the exact dynamic equations

$$\frac{\dot{k}}{k} = \frac{y - c - u - M\tau}{k} - \delta - n - \gamma, \quad (86a)$$

$$\frac{\dot{c}}{c} = r - \rho - \gamma, \quad (86b)$$

$$\frac{\dot{d}}{d} = n + \gamma - a, \quad (86c)$$

$$\frac{\dot{\tilde{q}}}{\tilde{q}} = r - \frac{x}{\tilde{q} B^2} - \eta \frac{a d \tau}{B} + a - n - \gamma, \quad (86d)$$

$$\dot{\tau} = -(n + \gamma)\tau. \quad (86e)$$

In particular, population growth cancels from the normalized consumption equation; it is not subtracted a second time.

Complete the terminal construction by setting

$$\bar{M} = \left[\frac{(n + \gamma) \bar{B}^{1-\eta}}{\chi} \right]^{1/\eta}, \quad \bar{\tilde{q}} = \frac{\bar{x}}{\bar{r} \bar{B}^2}, \quad \bar{d} = \frac{\bar{M}}{\eta(n + \gamma) \bar{\tilde{q}}}. \quad (87)$$

All are positive. Direct substitution shows that $(\bar{k}, \bar{c}, \bar{d}, \bar{\tilde{q}}, 0)$ is a fixed point of (86). This boundary point is a device for constructing paths with $\tau > 0$ and $B < \bar{B}$, not an admissible date-zero state itself.

I next verify saddle-path stability. Write $\mu = \eta/(1 - \eta)$, $r_k = \partial r / \partial \log k$, and $a_k = \partial[(y - u)/k] / \partial \log k + \bar{c}/\bar{k}$, evaluated at the fixed point. In $(\log k, \log c, \log d, \log \tilde{q}, \tau)$ coordinates the Jacobian has the two diagonal blocks

$$J_R = \begin{pmatrix} a_k & -\bar{c}/\bar{k} \\ r_k & 0 \end{pmatrix}, \quad J_B = \begin{pmatrix} -(n + \gamma)\mu & -(n + \gamma)\mu \\ (n + \gamma)\mu & \bar{r} + (n + \gamma)\mu \end{pmatrix}, \quad (88)$$

and the additional eigenvalue $-(n + \gamma)$. There can be forcing from τ to both blocks and from (k, c) to $(d, \tilde{q})$, but not in the reverse directions at the fixed point.

To check $r_k < 0$, let $s = s_X$, $u_\sigma = 1/\sigma$, $e = e_X$, and let $D(s)$ be the positive curvature expression in (65). Implicit differentiation of the monopoly condition gives

$$\frac{\partial \log X}{\partial \log K} = \frac{\alpha(1 - e)}{D(s)}, \quad r_k = (r + \delta)(1 - \alpha) \left[\frac{\alpha s(1 - e)}{D(s)} - 1 \right].$$

Moreover,

$$D(s) - \alpha s(1 - e) = (1 - s)\{u_\sigma(1 - e) + (\alpha - u_\sigma)(1 - u_\sigma)s\} > 0.$$

The sign is immediate when $u_\sigma \geq 1$ or $u_\sigma \leq \alpha$. For $\alpha < u_\sigma < 1$, the braces are linear in s , with positive endpoint values $u_\sigma(1 - u_\sigma)$ and $u_\sigma^2 + \alpha(1 - 2u_\sigma)$. The latter is positive if $u_\sigma \leq 1/2$; otherwise $\alpha < u_\sigma$ implies it exceeds $u_\sigma(1 - u_\sigma) > 0$. Thus $r_k < 0$ in every interior-share case.

Both blocks have negative determinants:

$$\det J_R = (\bar{c}/\bar{k})r_k < 0, \quad \det J_B = -(n + \gamma)\mu\bar{r} < 0.$$

There are therefore three stable and two unstable eigenvalues. The stable direction in J_R has a nonzero k component, and the stable direction in J_B has a nonzero d component. To verify that their number suffices to select the jump variables, consider a vector in the stable subspace whose state components (k, d, τ) are all zero. Its linearized τ path is zero. The homogeneous Ramsey block must then follow its stable direction; its zero initial k component forces its c component to be zero as well. With that forcing absent, the same argument in the research block forces $\tilde{q} = 0$. The projection of the three-dimensional stable subspace onto the three predetermined coordinates is therefore injective and hence invertible. This argument also covers repeated stable eigenvalues.

Lemma 1 makes the static solution smooth. The normalized vector field extends smoothly through $\tau = 0$, since $B = \bar{B} - d\tau$ remains positive nearby. The stable-manifold theorem and the inverse-function theorem therefore give a local graph $(c, \tilde{q})$ over (k, d, τ) ; see Dyatlov (2018). Choose a small neighborhood of the full fixed point and then a smaller initial-state neighborhood $\mathcal{U}_L$ of $(\bar{k}, \bar{d}, 0)$. Its constants depend on the parameters and the case under consideration. In original initial stocks the restriction is

$$(k_0, d_0, \tau_0) = \left(\frac{K_0}{A_0 N_0}, (\bar{B} - B_0)A_0 N_0, \frac{1}{A_0 N_0} \right) \in \mathcal{U}_L. \quad (89)$$

Closeness is required in all three coordinates; B_0 close to $\bar{B}$ alone is not sufficient. The point with $\tau = 0$ is a long-run limit, not a finite-date initial state. Every physical initial state satisfying (89) has a solution on this graph that stays in the full neighborhood for all future time and converges to the fixed point. It is unique among solutions with those properties. Positivity of $\tau, d, c, \tilde{q}, k$ and the original RSI technology ensure $0 < B_t < \bar{B}$ and $M_t > 0$ at every finite date. Reconstruction gives the original initial stocks, the equilibrium equations, and the limiting rates in (37)–(39). Indeed, the normalized vector field tends to zero at its fixed point, so the time derivatives of its smooth static ratios tend to zero as well. This step, not convergence of ratios alone, establishes convergence of the growth rates. Optimality and transversality are verified below.

Step 3. The AI-dominated case.

Let $\sigma > 1$ and $\bar{B} > \mathcal{B}(\sigma)$. Here $\varphi = (\sigma - 1)/\sigma \in (0, 1)$. Since capital outgrows effective labor, $K/(AN)$ has no finite limit. Instead normalize consumption and the shadow value by capital and

scale the efficiency gap by capital:

$$h = (AN/K)^\varphi, \quad d = (\bar{B} - B)K, \quad c = C/K, \quad v = q/K, \quad \tau = (AN)^{-1}.$$

Within this construction, y, x, u denote $Y/K, X/K, U/K$, respectively. The constant $\bar{d}$ now denotes the limit of $(\bar{B} - B)K$, not of $(\bar{B} - B)AN$; no original state or control is redefined. Then

$$B = \bar{B} - d\tau h^{1/\varphi}, \quad M = [vd\chi\eta B^\eta/\bar{B}]^{1/(1-\eta)}, \quad a = \chi B^\eta M^\eta/\bar{B}, \quad r = \alpha y - \delta.$$

Define $g_K = y - c - u - M\tau h^{1/\varphi} - \delta$. The exact equations are

$$\dot{h} = \varphi h(n + \gamma - g_K), \tag{90a}$$

$$\dot{d}/d = g_K - a, \tag{90b}$$

$$\dot{c}/c = n + r - \rho - g_K, \tag{90c}$$

$$\dot{v}/v = r - x/(vB^2) - \eta ad\tau h^{1/\varphi}/B + a - g_K, \tag{90d}$$

$$\dot{\tau} = -(n + \gamma)\tau. \tag{90e}$$

The CES term divided by capital is $[\omega_L h + \omega_X x^\varphi]^{1/\varphi}$. At $h = 0$ the monopoly equation has the strictly positive curvature $D(1) = \alpha(1 - \alpha)$. Its implicit static solution is therefore smooth in (h, B) near $(0, \bar{B})$. For the remaining terms, replacing $h^{1/\varphi}$ by $\max\{h, 0\}^{1/\varphi}$ outside the physical domain gives a C^1 extension because $1/\varphi > 1$. The C^1 stable-manifold theorem applies; see [Feldman \(2021, p. 7\)](#). The extension is only a mathematical device and does not change any equation on $h \geq 0$.

The limiting interest and aggregate growth rates are

$$\bar{r} = \mathcal{R}(\bar{B}), \quad \bar{g} = n + \bar{r} - \rho > n + \gamma. \tag{91}$$

At $h = \tau = 0$, the terminal constants are

$$\begin{aligned} \bar{y} &= \frac{\mathcal{R}(\bar{B}) + \delta}{\alpha}, & \bar{u} &= (1 - \alpha)^2 \bar{y}, & \bar{x} &= \bar{B} \bar{u}, \\ \bar{r} &= \mathcal{R}(\bar{B}), & \bar{g} &= n + \bar{r} - \rho, & \bar{c} &= \alpha(1 - \alpha)\bar{y} + \rho - n > 0, \\ \bar{v} &= \bar{x}/(\bar{r} \bar{B}^2), & \bar{M} &= [\bar{g} \bar{B}^{1-\eta}/\chi]^{1/\eta}, & \bar{d} &= \bar{M}/(\eta \bar{g} \bar{v}). \end{aligned} \tag{92}$$

The threshold implies $\bar{g} > n + \gamma > 0$ and $\bar{r} > \rho + \gamma > 0$, so all constants are positive. Substitution verifies the fixed point $(0, \bar{d}, \bar{c}, \bar{v}, 0)$.

The h and τ eigenvalues are $\varphi(n + \gamma - \bar{g}) < 0$ and $-(n + \gamma) < 0$. The first equals $\varphi[\rho + \gamma - \mathcal{R}(\bar{B})]$. It becomes zero at $\mathcal{R}(\bar{B}) = \rho + \gamma$, which is why this stable-manifold argument does not cover the threshold. The consumption-ratio eigenvalue is $\bar{c} > 0$. The remaining block in $(\log d, \log v)$ is

$$\begin{pmatrix} -\bar{g}\mu & -\bar{g}\mu \\ \bar{g}\mu & \bar{r} + \bar{g}\mu \end{pmatrix}, \quad \det = -\bar{g}\mu\bar{r} < 0.$$

The linearized system is triangular in the order $(h, \tau), c, (d, v)$. A stable vector with zero state components (h, d, τ) has zero h, τ paths; its consumption component must be zero because the unforced consumption equation is unstable. The research block then has zero d component and hence zero v component, as in the previous construction. Thus the three-dimensional stable subspace projects invertibly onto (h, d, τ) . The same graph argument gives a neighborhood $\mathcal{U}_A$ of $(0, \bar{d}, 0)$ such that the initial-state condition is

$$\left[\left(\frac{A_0 N_0}{K_0} \right)^\varphi, (\bar{B} - B_0) K_0, \frac{1}{A_0 N_0} \right] \in \mathcal{U}_A. \quad (93)$$

It constructs a convergent trajectory for every admissible initial state in this neighborhood, with a unique nearby convergent choice of C_0, q_0 and positive quantities at every finite date.

The normalized vector field tends to zero, so the time derivatives of the smooth positive ratios $Y/K, X/K$, and C/K tend to zero. Consequently $g_Y, g_X, g_C, g_K \rightarrow \bar{g}$. Also $s_X \rightarrow 1$ and $1 - s_X$ is asymptotically proportional to $(AN/X)^\varphi$. The wage equation gives

$$g_w \rightarrow \bar{g} - n + \varphi(n + \gamma - \bar{g}) = \gamma + \frac{\bar{r} - \rho - \gamma}{\sigma}.$$

This proves the growth and distribution limits in Proposition 1(iv).

Step 4. Optimality and transversality.

In either construction let $\bar{g}$ denote the terminal aggregate growth rate: $n + \gamma$ in the positive labor-share case and $n + \bar{r} - \rho$ in the AI-dominated case. The constructions imply

$$a \rightarrow \bar{g}, \quad M \rightarrow \left[\frac{\bar{g} \bar{B}^{1-\eta}}{\chi} \right]^{1/\eta} > 0, \quad g_q \rightarrow \bar{g}, \quad g_B \rightarrow 0, \quad M/Y \rightarrow 0.$$

Both transversality expressions in Definition 1 have limiting logarithmic growth

$$-\rho + n + g_K - g_C \rightarrow n - \rho < 0, \quad -r + g_q + g_B \rightarrow \bar{g} - \bar{r} = n - \rho < 0. \quad (94)$$

They therefore converge to zero, rather than merely remaining bounded.

Household utility is finite because $\log(C/N)$ grows at most linearly and $\rho > n$, as imposed in the household problem. Log utility is concave, the household budget is linear at given prices and distributions. More explicitly, let $\lambda_t = e^{-\rho t} N_t / C_t$, so $\dot{\lambda}_t = -r_t \lambda_t$ by (5). Concavity of log utility and equal initial capital give, for any admissible alternative $(\tilde{C}, \tilde{K})$,

$$\int_0^T e^{-\rho t} N_t \log \frac{\tilde{C}_t}{C_t} dt \leq \int_0^T \lambda_t (\tilde{C}_t - C_t) dt = -\lambda_T (\tilde{K}_T - K_T) \leq \lambda_T K_T \rightarrow 0.$$

Here $\tilde{K}_T \geq 0$ and the last limit is the household TVC. Final-good optimality follows from concavity and constant returns. The pointwise monopoly optimum is global by Lemma 1.

For the dynamic developer, feasibility and first-order conditions still require a sufficiency ar-

gument. Let $S_t = A_t N_t$ in the positive-share construction and $S_t = K_t$ in the AI-dominated construction, and let $D_t = \exp(-\int_0^t r_s ds)$. Homogeneity of the static problem and compactness of the normalized state neighborhood give a constant C_π such that $0 \leq \pi_t(b) \leq C_\pi S_t$ for every date and every $b \in [B_0, \bar{B}]$. Since $d \log(D_t S_t)/dt \rightarrow \bar{g} - \bar{r} = n - \rho < 0$, $\int_0^\infty D_t S_t dt < \infty$. Thus discounted operating profit is uniformly bounded above across all alternative research policies. The candidate's objective is finite, because its research compute is bounded and $r_t \rightarrow \bar{r} > 0$. An alternative with infinite discounted research expenditure has payoff $-\infty$ and cannot improve on it.

It remains to verify the curvature conditions (75) and (77) for every date and every alternative AI-efficiency level in $[B_0, \bar{B})$, not just on the selected path. First fix a compact normalized-state neighborhood and an interval of counterfactual efficiencies $[b_*, \bar{B}]$, with $0 < b_* < \bar{B}$. In the positive labor-share construction, $K/(AN)$ stays in a compact neighborhood of $\bar{k} > 0$. Smoothness of the static monopoly solution bounds $\varepsilon_{B,t}(b)$ uniformly for these states and for AI-efficiency levels near $\bar{B}$. In the AI-dominated construction, h stays near zero, and the same uniform bound follows from the smooth static limit $\varepsilon_{B,t}(b) \rightarrow 1/\alpha$. Thus in either case there is a finite constant E with $\varepsilon_{B,t}(b) - 2 \leq E$ throughout that state domain. This bound is chosen before shrinking the initial-state neighborhood. Because $\mu = \eta/(1 - \eta)$ is finite for each $0 < \eta < 1$, choose $b_{**} \in [b_*, \bar{B})$ sufficiently close to $\bar{B}$ that

$$(1 - b_{**}/\bar{B}) \max\{E, \mu - 1, 0\} < b_{**}/\bar{B}.$$

Shrink the open neighborhood $\mathcal{U}_L$ or $\mathcal{U}_A$ so that $B_0 > b_{**}$. This is possible because $B_0 = \bar{B} - d_0 \tau_0$ in the first construction and $B_0 = \bar{B} - d_0 \tau_0 h_0^{1/\varphi}$ in the second. Every feasible research policy has $b \geq B_0$, so the two curvature conditions hold throughout the entire alternative-state domain at every date. Lemma 2 and the verified TVC therefore establish global developer optimality without imposing $\eta \leq 1/2$. No curvature property of the candidate is being assumed: it follows from the primitives and the chosen initial-state neighborhood.

All quantities remain finite at every finite date, and market clearing and the original initial conditions hold by reconstruction. The paths are therefore equilibria, not merely solutions of the canonical equations. The result is local only in its set of initial stocks. For $\sigma > 1$, at $\bar{B} = \mathcal{B}(\sigma)$ the interior-share terminal point reaches the boundary and a stable eigenvalue in the boundary normalization becomes zero; this proof does not apply. For $\sigma < 1$ and $\bar{B} \leq \mathcal{B}(\sigma)$, it does not rule out equilibria with different long-run ratios. Nor does it imply existence at arbitrary initial stocks or in the uncapped limit.

Step 5. Initial-state sets and the four conclusions.

The preimage of each open neighborhood under its continuous initial-state map, restricted to $A_0, N_0, K_0 > 0$ and $0 < B_0 < \bar{B}$, is open in the original initial stocks. It is nonempty. In the positive-labor-share cases, choose k_0, d_0 close to $\bar{k}, \bar{d}$ and $\tau_0 > 0$ sufficiently small. Then $A_0 N_0 = 1/\tau_0$, $K_0 = k_0/\tau_0$, and $B_0 = \bar{B} - d_0 \tau_0$ are admissible. In the AI-dominated case, choose d_0 near $\bar{d}$ and $h_0, \tau_0 > 0$ sufficiently small. The inverse relations are $A_0 N_0 = 1/\tau_0$, $K_0 = 1/(\tau_0 h_0^{1/\varphi})$,

and $B_0 = \bar{B} - d_0 \tau_0 h_0^{1/\varphi}$. Any positive factorization of $A_0 N_0$ supplies A_0, N_0 . These choices also satisfy the strict efficiency restriction in Step 4.

For each initial state in the resulting set, the local graph selects $C_0, q_0 > 0$. These jump values are unique among the trajectories that remain near the indicated normalized limit and converge to it. The argument does not assert global uniqueness.

For $\sigma < 1$ with $\bar{B} > \mathcal{B}(\sigma)$, and for $\sigma > 1$ with $\bar{B} < \mathcal{B}(\sigma)$, Step 1 gives a unique interior $\bar{s}_X$. Step 2 then yields $g_Y - n \rightarrow \gamma$, $g_w \rightarrow \gamma$, and $r \rightarrow \rho + \gamma$. At $\sigma = 1$, the exact CES limit gives $s_X = \omega_X$ at every date, and the same construction works for every finite positive upper bound. These establish the first three cases of the proposition. Step 3 gives $s_X \rightarrow 1$, $r \rightarrow \mathcal{R}(\bar{B})$, $g_Y - n \rightarrow \mathcal{R}(\bar{B}) - \rho$, and the wage-growth limit in (41), establishing the fourth case. Step 4 verifies the agent-optimality and transversality requirements of Definition 1 for all these trajectories. $\square$

Proof of the AI-bottleneck statement following Proposition 1. Consider any equilibrium with $0 < \sigma < 1$ and $0 < B_0 < \bar{B} < \mathcal{B}(\sigma)$. The increasing share map in equation (81), established in Step 1 above, and $B_t \leq \bar{B}$ imply

$$r_t \leq \mathcal{R}(B_t) \leq \mathcal{R}(\bar{B}) < \rho + \gamma, \quad \frac{Y_t}{K_t} \leq \frac{\mathcal{R}(\bar{B}) + \delta}{\alpha}. \quad (95)$$

The developer's discounted continuation net profit is nonnegative: it can stop research and earn nonnegative operating profit. Wage income is also nonnegative. Integrating the household budget (4), using its transversality condition (6) and the Euler equation (5), therefore gives the first inequality below; integrating the Euler equation with the return bound gives the second:

$$C_t \geq (\rho - n)K_t, \quad \frac{C_t}{N_t} \leq \frac{C_0}{N_0} e^{[\mathcal{R}(\bar{B}) - \rho]t}. \quad (96)$$

Combining these inequalities yields

$$\frac{Y_t}{N_t} \leq \frac{\mathcal{R}(\bar{B}) + \delta}{\alpha(\rho - n)} \frac{C_0}{N_0} e^{[\mathcal{R}(\bar{B}) - \rho]t}, \quad \limsup_{t \rightarrow \infty} \frac{1}{t} \log \left(\frac{Y_t/N_t}{Y_0/N_0} \right) \leq \mathcal{R}(\bar{B}) - \rho < \gamma. \quad (97)$$

Consequently $Y/(AL) \rightarrow 0$. Since equation (32) gives $X/Y \leq \bar{B}(1 - \alpha)^2$ under complementarity, $X/(AL) \rightarrow 0$: AI production services become scarce relative to effective labor. Equation (10) then implies $s_X \rightarrow 1$ and hence $wL/Y \rightarrow 0$. The growth bound concerns the long-run average, not convergence of the instantaneous growth rate. Neither convergence of C/Y nor $B_t \rightarrow \bar{B}$ is assumed. These are necessary properties of any equilibrium in this region, not a proof of equilibrium existence. $\square$

Research expenditure and finite-horizon value. Besides the two cases in Proposition 5, the proof gives auxiliary finite-horizon bounds for the same given paths. At $\eta = \alpha$, the leading benefit and cost terms of the research-burst construction below have the same order, so comparing exponents alone does not settle the result. With a fixed finite upper bound, the truncated problem has a finite value and attains its maximum for every $0 < \eta < 1$. Global verification of a candidate research policy is addressed separately in Lemma 2.

Proof of Proposition 5. Part (i). I first consider the uncapped economy. Define

$$\theta = \frac{1 - \alpha}{\alpha}, \quad S = \int_0^T M_t dt.$$

Here S is total research expenditure over the fixed horizon $[0, T]$. For an admissible research policy, let its discounted net payoff be

$$J_T(M) = \int_0^T \exp \left\{ - \int_0^t r_s ds \right\} [\pi(B_t) - M_t] dt.$$

For every admissible policy, B is nondecreasing. The RSI technology implies

$$\frac{d}{dt} B_t^{1-\eta} = (1 - \eta) \chi M_t^\eta.$$

Integration and Hölder's inequality give

$$\begin{aligned} B_t^{1-\eta} &= B_0^{1-\eta} + (1 - \eta) \chi \int_0^t M_s^\eta ds, \\ \int_0^T M_t^\eta dt &\leq T^{1-\eta} S^\eta. \end{aligned}$$

Because $0 < \eta < 1$, spreading a given expenditure evenly over $[0, T]$ maximizes the integral of M^η . Hence there is a constant $C_B > 0$, independent of the research policy, such that

$$\max_{0 \leq t \leq T} B_t \leq C_B \left(1 + S^{\eta/(1-\eta)} \right), \quad (98)$$

The same bound covers arbitrarily concentrated controls because, on a support of length $d \leq T$, $\int M_t^\eta dt \leq d^{1-\eta} S^\eta$.

Let $\bar{K} = \max_{t \leq T} K_t$ and $\bar{\mathcal{L}} = \max_{t \leq T} A_t L_t$. The weighted power-mean inequality gives $Z \leq AL + X$. The final-good firm's demand condition therefore implies

$$p_X X \leq (1 - \alpha) \bar{K}^\alpha \left(\bar{\mathcal{L}}^{1-\alpha} + X^{1-\alpha} \right).$$

Write the right-hand side as $a + bX^{1-\alpha}$, with $a, b > 0$ determined by the given paths and production parameters. The first-order condition for maximizing this revenue bound net of X/B is $b(1-\alpha)X^{-\alpha} = 1/B$, and its maximized value is

$$\max_{X \geq 0} \left\{ a + bX^{1-\alpha} - \frac{X}{B} \right\} = a + \alpha b [b(1-\alpha)B]^\theta.$$

There is therefore a constant $C_\pi > 0$, independent of the research policy, such that

$$\pi(B) \leq C_\pi(1 + B^\theta). \quad (99)$$

Combining the AI-efficiency and profit bounds shows that operating profit grows no faster than a constant plus a term proportional to

$$S^{\eta(1-\alpha)/[\alpha(1-\eta)]}, \quad (100)$$

where the exponent is the product of $\eta/(1-\eta)$, from expenditure to AI efficiency, and $(1-\alpha)/\alpha$, from AI efficiency to profit. This is an upper bound on the order of growth, not an exact profit function.

On a finite horizon, integrability of r bounds the discount factor above and away from zero. Let $\underline{D} > 0$ be its minimum on $[0, T]$. Combining (98) and (99) yields, for constants $C_0, C_1 > 0$ independent of the research policy,

$$J_T(M) \leq C_0 + C_1 S^p - \underline{D}S, \quad p = \frac{\eta(1-\alpha)}{\alpha(1-\eta)}. \quad (101)$$

The exponent satisfies $p < 1$ exactly when $\eta < \alpha$. In this case, the benefit grows at most sublinearly in expenditure while its cost grows linearly. The bound tends to minus infinity as $S \rightarrow \infty$, so the value is finite and the objective is coercive.

Part (ii). An upper bound with $p > 1$ does not by itself prove unboundedness. For the converse, I construct research policies with unbounded payoffs. Suppose $\sigma > 1$. As $B \rightarrow \infty$, optimized operating profit satisfies

$$\pi(B) \sim \kappa_X K B^\theta$$

for a constant $\kappa_X > 0$, uniformly when K, A , and L remain in compact subsets of the positive orthant. Fix $\Delta \in (0, T)$, choose $M = \mathcal{M}$ on $[0, \Delta]$, and choose $M = 0$ afterward. Then

$$B_{\mathcal{M}}(t)^{1-\eta} = B_0^{1-\eta} + (1-\eta)\chi \mathcal{M}^\eta \min\{t, \Delta\}.$$

Choosing X optimally at every date gives

$$J_T(\mathcal{M}) = c_\pi \mathcal{M}^{\eta(1-\alpha)/[\alpha(1-\eta)]} - c_M \mathcal{M} + o\left(\mathcal{M}^{\eta(1-\alpha)/[\alpha(1-\eta)]}\right), \quad (102)$$

with $c_\pi, c_M > 0$. The exponent exceeds one when $\eta > \alpha$, so this family makes the truncated value unbounded above. At equality, the exponent is one and the sign of the leading coefficient

determines this family's result.

To obtain the infinite-horizon nonexistence conclusion, suppose instead that an uncapped equilibrium has a developer policy $\widehat{M}$, an associated efficiency path $\widehat{B}$, and a finite discounted value $\widehat{V}_0$. On every finite interval of this candidate equilibrium, K, A, L are positive and continuous, and $r = \alpha Y/K - \delta$ is continuous by the static production and monopoly solution. The finite-horizon argument therefore applies to these paths, without requiring a BGP or a bounded interest rate as $t \rightarrow \infty$. Hold its paths of K, A, L , and r fixed, as required by the developer's problem (21). On $[0, T]$, use the burst policy above; after T , resume $\widehat{M}$. For a sufficiently large burst, $B_{\mathcal{M}}(T) > \widehat{B}(T)$. Under the uncapped law, the difference $B_{\mathcal{M}}(t)^{1-\eta} - \widehat{B}(t)^{1-\eta}$ is then constant and positive for $t \geq T$, since the two policies use the same research spending. Operating profit is increasing in B by equation (20), so subsequent net profit is no lower than under the candidate policy. Let $\widehat{J}_T$ be the candidate's discounted net profit on $[0, T]$ and let $J_{\infty}(\mathcal{M})$ denote the payoff from the deviating policy, including its continuation. Then

$$J_{\infty}(\mathcal{M}) \geq J_T(\mathcal{M}) + \widehat{V}_0 - \widehat{J}_T \rightarrow +\infty.$$

The continuation payoff is well defined as a finite value or $+\infty$: its difference from the candidate's continuation payoff is the integral of a nonnegative operating-profit gain. Each deviation has finite research expenditure on every finite interval and hence finite B there by (98). Every finite-valued candidate can therefore be improved, contradicting developer optimality. The argument excludes a finite-valued optimum; it does not treat an infinite payoff as an attained firm value or assume that a trajectory reaches a singularity.

For a finite upper bound, the invariance result above confines B to the compact interval $[B_0, \overline{B}]$. Optimized operating profit is continuous in B and uniformly bounded on the fixed horizon. Hence $J_T(M) \leq C_0 - \underline{D}S$, so every maximizing sequence has bounded total expenditure. Set $v = M^{\eta}$ and $p = 1/\eta > 1$. The control cost is v^p , while the state equation is affine in v :

$$\dot{B} = \chi B^{\eta} \left(1 - \frac{B}{\overline{B}}\right) v.$$

A maximizing sequence is bounded in the reflexive space $L^p[0, T]$. A weakly convergent subsequence and the associated uniformly convergent state paths preserve feasibility. The operating-profit integral is continuous under uniform state convergence, while $\int v^p$ is weakly lower semicontinuous. The objective is therefore weakly upper semicontinuous and attains its maximum. $\square$

The condition $\eta < \alpha$ prevents arbitrarily large research expenditure from being optimal on a fixed horizon in the uncapped economy. It is a sufficient restriction, not a necessary one for every value of σ , and finite-horizon existence with a finite upper bound does not require $\eta < \alpha$. Neither the bound for $\eta < \alpha$ nor finite-horizon developer existence with a finite upper bound establishes an equilibrium: the paths of K, A, L , and r were taken as given. In contrast, for $\sigma > 1$ and $\eta > \alpha$, the profitable-deviation argument rules out an uncapped equilibrium with a finite-valued developer

optimum at any such candidate paths. The quantitative analysis uses $\eta = 0.20 < \alpha = 0.33$, for which the expenditure exponent is approximately 0.508; the value of η is illustrative rather than estimated.

Proof of Proposition 2. Equations (43) and (44) follow because $\sigma/(\sigma-1) < 0$. Define $k = K/(AL)$. The resource constraint (29a), nonnegative consumption and compute expenditure, and $L = N$ imply

$$\dot{k} \leq \omega_L^{\sigma(1-\alpha)/(\sigma-1)} k^\alpha - (\delta + n + \gamma)k. \quad (103)$$

Since $\delta + n + \gamma > 0$ and $0 < \alpha < 1$, the right-hand side is negative above a finite positive level. In particular,

$$k_t \leq \max \left\{ k_0, \left[\frac{\omega_L^{\sigma(1-\alpha)/(\sigma-1)}}{\delta + n + \gamma} \right]^{1/(1-\alpha)} \right\}.$$

Equation (44) then bounds $Y/(AL)$ uniformly. Since $A_t = A_0 e^{\gamma t}$ and $L = N$, Y/N is bounded above by a constant times $e^{\gamma t}$. Taking logarithms, dividing by t , and taking the upper limit gives (45).

The same bounds keep K and Y finite on every bounded time interval. For any such horizon T , integrating the resource constraint gives

$$\int_0^T M_t dt \leq K_0 + \int_0^T Y_t dt < \infty.$$

The uncapped RSI technology and Hölder's inequality imply

$$\begin{aligned} B_T^{1-\eta} &= B_0^{1-\eta} + (1-\eta)\chi \int_0^T M_t^\eta dt \\ &\leq B_0^{1-\eta} + (1-\eta)\chi T^{1-\eta} \left(\int_0^T M_t dt \right)^\eta < \infty. \end{aligned} \quad (104)$$

Thus B cannot diverge at a finite date either. This argument bounds stocks and output; it is not a regularity theorem for all instantaneous control choices or an equilibrium-existence proof.

There is also no finite-horizon unbounded-payoff argument of the kind used for substitution in Proposition 5. At given positive continuous paths of K, A, L on $[0, T]$, optimized operating profit satisfies

$$0 \leq \pi_t(B) \leq p_X X \leq (1-\alpha)Y \leq (1-\alpha)\omega_L^{\sigma(1-\alpha)/(\sigma-1)} K_t^\alpha (A_t L_t)^{1-\alpha}.$$

This upper bound is independent of the developer's choice of B . For a given locally integrable r , the discount factor $D_t = \exp(-\int_0^t r_s ds)$ has a positive minimum on $[0, T]$. Discounted operating revenue is bounded, whereas the cost of research tends to infinity with $\int_0^T M_t dt$. The developer's truncated supremum is therefore finite, and net payoff tends to minus infinity as total research expenditure diverges. This conclusion holds for every $0 < \eta < 1$, but does not prove that an infinite-horizon optimum exists. $\square$

Proof of Proposition 3. All auxiliary variables below are confined to this proof. Write $x = X/(AL)$, $k = K/(AL)$, $c = C/(AL)$, $y = Y/(AL)$, $u = U/(AL)$, $m = M/(AL)$, $d = \rho - n > 0$, and $\nu = \delta + n + \gamma$. No convergence of a growth rate or allocation ratio is assumed.

Step 1. Static optimality gives a limiting production function. Put $\varphi = (\sigma - 1)/\sigma < 0$ and $f(x) = [\omega_L + \omega_X x^\varphi]^{(1-\alpha)/\varphi}$, so $y = k^\alpha f(x)$. The monopoly condition (19) requires $e_X = (1 - s_X)/\sigma + \alpha s_X < 1$. Thus

$$s_X > s_\infty := \frac{1 - \sigma}{1 - \alpha\sigma}, \quad 0 < x < x_\infty := \left[\frac{\omega_L s_\infty}{\omega_X (1 - s_\infty)} \right]^{1/\varphi}. \quad (105)$$

Define $f_\infty = f(x_\infty) > 0$. In particular, $y \leq f_\infty k^\alpha$ and $0 < u \leq x_\infty/B$.

The static condition can be written

$$k^\alpha v(x) = 1/B, \quad v(x) = \frac{(1 - \alpha)s_X f(x)(1 - e_X)}{x}.$$

On $(0, x_\infty)$, v is strictly decreasing, from infinity to zero. Indeed, with a prime here denoting differentiation with respect to $\log x$, $e'_X = (\alpha - \sigma^{-1})\varphi s_X(1 - s_X) > 0$ and $-d \log v / d \log x = e_X + e'_X/(1 - e_X) > \alpha s_X$. Consequently x increases with B and k , while

$$\frac{\partial \log y}{\partial \log k} = \alpha + \frac{\alpha(1 - \alpha)s_X}{e_X + e'_X/(1 - e_X)} < 1.$$

Thus y/k decreases with k and increases with B . For k in any compact interval bounded away from zero, as $B \rightarrow \infty$,

$$x \rightarrow x_\infty, \quad y \rightarrow f_\infty k^\alpha, \quad u \rightarrow 0$$

uniformly. These are properties of the static map, not assumptions about the equilibrium path. They also imply that, for any prescribed finite return, sufficiently small k and sufficiently large B make r exceed that return, uniformly over those small values of k .

Step 2. Household and developer optimality bound consumption. Proposition 2 bounds k and y above by finite constants $k_{\max}, y_{\max}$. Euler and $r \geq -\delta$ give $\dot{c}/c \geq -(\delta + \rho + \gamma)$. Integrating the resource constraint over $[t, t + 1]$ therefore gives

$$c_t \int_0^1 e^{-(\delta + \rho + \gamma)v} dv \leq \int_t^{t+1} c_s ds \leq k_{\max} + y_{\max}.$$

Hence c has a finite uniform upper bound $c_{\max}$.

Write $D_{t,s} = \exp(-\int_t^s r_v dv)$. Euler implies $\int_t^\infty D_{t,s} C_s ds = C_t/d$. Integrating the household budget (4) and using its TVC (6) shows that the discounted sum of wages and distributions has a finite limit. Since wages are nonnegative, the discounted distributions alone have a limit that is either finite or minus infinity. The latter is impossible: the developer can stop research and obtain nonnegative operating profits. By continuation optimality, its remaining value $V_t = \int_t^\infty D_{t,s} \Pi_s ds$

is therefore finite and nonnegative. It follows that

$$\frac{C_t}{d} = K_t + V_t + \int_t^\infty D_{t,s} w_s L_s ds, \quad c_t \geq dk_t, \quad \int_t^\infty D_{t,s} w_s L_s ds \leq \frac{C_t}{d}. \quad (106)$$

Here V_t is an auxiliary continuation value, not an additional traded asset or a change to the household budget.

Step 3. Future research spending is small before taking growth limits. Fix a positive rate a . Step 1 permits a small $k_a > 0$ and a sufficiently late date such that $r - (n + \gamma) \geq a$ whenever $k \leq k_a$. On $[k_a, k_{\max}]$, the wage equation (12) and the static map give a uniform positive lower bound ℓ_a on w/A . Since $r \geq -\delta$, there is a finite constant $H_a > 0$ such that

$$a \leq r - (n + \gamma) + H_a w/A$$

at every sufficiently late date. Multiplication by $D_{t,s} A_s L_s$, integration, and (106) give

$$\int_t^\infty D_{t,s} A_s L_s ds \leq \frac{A_t L_t + H_a C_t/d}{a} \leq J A_t L_t \quad (107)$$

for a fixed finite J . The omitted terminal term is nonpositive; no discount-rate limit is used.

The research condition (22) and costate equation (23) imply

$$\frac{d(qB)}{dt} = r q B - U + \frac{1 - \eta}{\eta} M. \quad (108)$$

Since B is nondecreasing, $U_s \leq x_\infty A_s L_s / B_t$ for $s \geq t$. Integrating (108) first over finite intervals and then using the developer TVC (25) gives

$$q_t B_t + \frac{1 - \eta}{\eta} \int_t^\infty D_{t,s} M_s ds = \int_t^\infty D_{t,s} U_s ds \leq \frac{x_\infty^J}{B_t} A_t L_t. \quad (109)$$

The right side is finite by (107); nonnegativity then ensures that the research integral is finite as well. Its value divided by $A_t L_t$ tends to zero. This argument requires $0 < \eta < 1$, but neither $\eta \leq 1/2$ nor $\eta < \alpha$.

Step 4. Capital cannot repeatedly approach zero. Let $F(k) = f_\infty k^\alpha - \nu k$. Choose $\varepsilon > 0$ small enough that F is positive and increasing on $(0, \varepsilon]$. Step 1 also permits a sufficiently late date and $a > 0$ such that $\dot{c}/c = r - \rho - \gamma \geq a$ whenever $k \leq \varepsilon$. For any $j \leq \varepsilon$, a feasible path cannot enter $\{k \leq j, c \geq F(j)\}$: there $\dot{k} \leq F(k) - c \leq 0$, whereas c continues to grow at least at rate a , contradicting its upper bound.

If k stayed below ε forever, consumption would likewise exceed its upper bound. Hence the path eventually reaches ε . Suppose arbitrarily late crossings from ε to $\varepsilon/2$ were possible. Let t_0 be the last crossing of ε before the first subsequent crossing t_1 of $\varepsilon/2$. Throughout this interval

$k \in [\varepsilon/2, \varepsilon]$ and c increases. Equation (106) gives $c_{t_0} \geq d\varepsilon$, so

$$t_1 - t_0 \leq T_\varepsilon := a^{-1} \log\left(\frac{c_{\max}}{d\varepsilon}\right) < \infty.$$

At t_1 , feasibility requires $c_{t_1} < F(\varepsilon/2)$ by the preceding invariant-region argument. Thus $c_s < F(\varepsilon/2)$ throughout the crossing. Uniform static convergence on the crossing interval yields numbers $\epsilon_{t_0} \rightarrow 0$ such that

$$\dot{k}_s = y_s - u_s - \nu k_s - c_s - m_s \geq -\epsilon_{t_0} - m_s.$$

On this interval r is bounded above by a fixed positive R_ε . Since $n + \gamma > 0$, equation (109) then implies

$$\frac{\varepsilon}{2} \leq T_\varepsilon \epsilon_{t_0} + \int_{t_0}^{t_1} m_s ds \leq T_\varepsilon \epsilon_{t_0} + e^{R_\varepsilon T_\varepsilon} \frac{\eta x_\infty J}{(1-\eta)B_{t_0}} \rightarrow 0,$$

a contradiction. Therefore k is eventually bounded away from zero, and so is $c \geq dk$. In particular, r is now bounded above.

Step 5. Research vanishes pointwise and Ramsey dynamics determine the limit. Differentiating the research condition and using the costate equation gives

$$(1-\eta)g_M = r - \frac{\eta \chi U B^{\eta-1}}{M^{1-\eta}} \leq r. \quad (110)$$

Choose a positive R above all eventual values of r and put $H = R/(1-\eta)$. Then $M_s \geq M_t e^{-H(t-s)}$ on $[t-1, t]$ at sufficiently late dates. Consequently

$$\frac{1}{A_{t-1}L_{t-1}} \int_{t-1}^t D_{t-1,s} M_s ds \geq e^{n+\gamma-R-H} m_t.$$

The left side tends to zero by (109). Hence $m \rightarrow 0$, and Step 1 already gives $u \rightarrow 0$.

The normalized equilibrium equations therefore approach, uniformly on the compact positive region containing the path, the Ramsey system

$$\dot{k} = F(k) - c, \quad \dot{c} = c[\alpha f_\infty k^{\alpha-1} - \delta - \rho - \gamma]. \quad (111)$$

For clarity, the following argument verifies convergence rather than inferring it solely from convergence of the equations. This limiting system has one positive stationary point,

$$k_\infty = \left[\frac{\alpha f_\infty}{\rho + \gamma + \delta} \right]^{1/(1-\alpha)}, \quad c_\infty = F(k_\infty) > 0.$$

Its Jacobian has negative determinant, so it is a saddle. The standard Ramsey phase portrait is discussed by Groth (2013). There is no nonconstant complete orbit confined to a compact subset

of the positive quadrant. To see this, the multiplier $1/c^2$ gives

$$\frac{\partial}{\partial k} \frac{F(k) - c}{c^2} + \frac{\partial}{\partial c} \frac{\alpha f_\infty k^{\alpha-1} - \delta - \rho - \gamma}{c} = \frac{\rho - n}{c^2} > 0.$$

The Bendixson–Dulac argument excludes both periodic orbits and homoclinic loops. The planar limit-set theorem, the unique stationary point, and its saddle character then exclude a nonconstant complete orbit confined to such a compact set: its forward and backward limits would otherwise require a periodic orbit or a connection back to that same saddle.

For any sequence of dates tending to infinity, time translations of our bounded positive path have a subsequence converging on every bounded time interval to a complete orbit of (111). This follows from bounded derivatives, uniform convergence of the vector fields, and the integral form of the differential equations. The preceding paragraph forces that orbit to be the stationary point. Thus every accumulation point is (k_∞, c_∞) , proving convergence of the original path.

It follows that $y \rightarrow y_\infty = f_\infty k_\infty^\alpha > 0$ and

$$r \rightarrow \rho + \gamma, \quad \frac{K}{Y} \rightarrow \frac{\alpha}{\rho + \gamma + \delta}, \quad \frac{C}{Y} \rightarrow 1 - \frac{\alpha(\delta + n + \gamma)}{\rho + \gamma + \delta} > 0. \quad (112)$$

Positivity follows because the numerator of the final expression is $\rho - n + (1 - \alpha)(\delta + n + \gamma) > 0$. This proves (46). Since $u, m \rightarrow 0$ and $y \rightarrow y_\infty > 0$, it follows that $U/Y = u/y \rightarrow 0$ and $M/Y = m/y \rightarrow 0$, proving (47). It also gives $\dot{k} \rightarrow 0$ and $g_C \rightarrow n + \gamma$.

Step 6. Output and wage growth converge as well. A convergent level need not have a convergent derivative. The bound on g_M from Step 5 and the RSI technology give

$$B_t^{1-\eta} \geq (1 - \eta) \chi M_t^\eta \int_0^1 e^{-\eta H v} dv.$$

Thus $g_B = \chi M^\eta / B^{1-\eta}$ is bounded above. At x_∞ , the marginal-revenue function v defined in Step 1 has a simple zero: $v(x_\infty) = 0$ and $v'(x_\infty) < 0$, where the prime now denotes differentiation with respect to x . Differentiating $k^\alpha v(x) = 1/B$ yields

$$\dot{x} = -\frac{v(x)}{v'(x)} \left(g_B + \alpha \frac{\dot{k}}{k} \right) \rightarrow 0.$$

Hence $\dot{y} \rightarrow 0$ and $\dot{s}_X \rightarrow 0$. Since $y_\infty > 0$ and $1 - s_\infty > 0$, $g_{Y/N} \rightarrow \gamma$ and, by (12), $g_w \rightarrow \gamma$. The limiting income shares follow by substituting s_∞ into (12) and (13). The result is conditional on equilibrium existence and $B \rightarrow \infty$, not an equilibrium-existence theorem or an assumption of balanced growth. $\square$

Proof of Proposition 4. Use the definitions in Appendix C. *The reference BGP.* First, all candidate ratios are positive. The research denominator $\rho - n + \eta g_Y^*$ is positive. Moreover,

$$i^* = \frac{\alpha(g_Y^* + \delta)}{\rho - n + g_Y^* + \delta} < \alpha, \quad m^* < \frac{u^* \eta}{1 - \eta}.$$

Since $\Delta = (1 - \alpha)(1 - \eta - \omega_X) > 0$, $\omega_X < 1 - \eta$ and $\omega_X < 1$. Consequently,

$$u^* + m^* < \frac{u^*}{1 - \eta} < (1 - \alpha)\omega_X < 1 - \alpha, \quad c^* = 1 - u^* - i^* - m^* > 0.$$

Thus positive consumption follows from the stated conditions; it is not an additional assumption.

For given $A_0, N_0 > 0$, choose the reference initial stocks as

$$B_0^* = \left[A_0 N_0 (k^*)^{\alpha/[(1-\alpha)\omega_L]} (u^*)^{\omega_X/\omega_L} m^* \left(\frac{\chi}{g_B^*} \right)^{1/\eta} \right]^{\eta\omega_L/(1-\eta-\omega_X)}, \quad (113)$$

$$Y_0^* = A_0 N_0 (k^*)^{\alpha/[(1-\alpha)\omega_L]} (u^* B_0^*)^{\omega_X/\omega_L}, \quad K_0^* = k^* Y_0^*. \quad (114)$$

The exponent is positive. These choices solve simultaneously the initial production equation and $g_B^* B_0^* = \chi(B_0^* m^* Y_0^*)^\eta$. Set $U_0^* = u^* Y_0^*$, $M_0^* = m^* Y_0^*$, $C_0^* = c^* Y_0^*$, $X_0^* = B_0^* U_0^*$, and

$$q_0^* = \frac{M_0^*}{\eta g_B^* B_0^*}. \quad (115)$$

Let $Y_t^* = Y_0^* e^{g_Y^* t}$, $B_t^* = B_0^* e^{g_B^* t}$, $K_t^* = k^* Y_t^*$, $C_t^* = c^* Y_t^*$, $U_t^* = u^* Y_t^*$, $M_t^* = m^* Y_t^*$, and $q_t^* = q_0^* e^{(g_Y^* - g_B^*)t}$. Use the exogenous A_t, N_t and set $L_t = N_t$ and $X_t^* = B_t^* U_t^*$. The static equilibrium equations determine all remaining quantities and prices.

The two growth identities (52) and (53) ensure that the production equation and the RSI technology hold at every date once they hold initially. The static monopoly condition gives $p_X X = (1 - \alpha)\omega_X Y$ and $U = u^* Y$. Equation (115) gives $M = \eta q \dot{B}$, which is the research first-order condition (28k) with $\psi = 1$. The definitions of k^* , r^* , and c^* verify the capital-return, Euler, and resource equations. The costate equation reduces to

$$g_Y^* - g_B^* = r^* - \frac{u^* \eta g_B^*}{m^*} - \eta g_B^*.$$

This holds by equation (139) and $(1 - \eta)g_B^* = \eta g_Y^*$. All allocations are positive and finite at every finite date.

Both transversality conditions are explicit:

$$e^{-\rho t} \frac{N_t K_t^*}{C_t^*} = \frac{N_0 k^*}{c^*} e^{-(\rho-n)t} \longrightarrow 0, \quad (116)$$

$$e^{-r^* t} q_t^* B_t^* = q_0^* B_0^* e^{-(\rho-n)t} \longrightarrow 0. \quad (117)$$

Discounted developer profit is a constant multiple of $e^{-(\rho-n)t}$. Discounted household utility is $N_0 e^{-(\rho-n)t} [\log(C_0^*/N_0) + (g_Y^* - n)t]$. Both improper objective integrals are finite.

It remains to establish global optimality, rather than only the dated necessary conditions. At given paths of K, A, L , static optimization gives

$$\begin{aligned} \pi_t(B) &= (1-\alpha)\omega_X [1 - (1-\alpha)\omega_X] [K_t^\alpha (A_t L_t)^{(1-\alpha)\omega_L}]^{1/[1-(1-\alpha)\omega_X]} \\ &\quad \times [(1-\alpha)^2 \omega_X^2 B]^{(1-\alpha)\omega_X/[1-(1-\alpha)\omega_X]}. \end{aligned} \quad (118)$$

First suppose $\eta \leq 1/2$. This function is concave in B when $(1-\alpha)\omega_X \leq 1/2$. The RSI technology $F(B, M) = \chi B^\eta M^\eta$ is jointly concave when $\eta \leq 1/2$. Since $q > 0$, the developer's current-value Hamiltonian $\pi_t(B) - M + q_t F(B, M)$ is jointly concave, and its first-order conditions maximize it globally. Write $D_t = \exp(-\int_0^t r_s ds)$ and $J_T = \int_0^T D_t [\pi_t(B_t) - M_t] dt$. For any admissible alternative with the same initial B , the supporting hyperplane inequality and the costate equation imply

$$\tilde{J}_T - J_T \leq -D_T q_T (\tilde{B}_T - B_T) \leq D_T q_T B_T \rightarrow 0. \quad (119)$$

Hence no admissible alternative has a greater limiting payoff, or a greater upper limit of truncated payoff, than this finite-valued candidate. For $\eta > 1/2$, use only within this proof the invertible state transformation and its current-value costate

$$S = B^{1-\eta}, \quad \dot{S} = (1-\eta)\chi M^\eta, \quad p = \frac{qB^\eta}{1-\eta}. \quad (120)$$

This is the same feasible research problem, not a different technology. At given K, A, L , equation (118) becomes a positive time-dependent coefficient times

$$S^{\frac{(1-\alpha)\omega_X}{[1-(1-\alpha)\omega_X](1-\eta)}}.$$

It is concave in S if $(1-\alpha)\omega_X \leq (1-\eta)/(2-\eta)$. The maintained restrictions imply

$$(1-\alpha)\omega_X < (1-\eta)^2 < \frac{1-\eta}{2-\eta} \quad (\eta > 1/2).$$

Consequently $\pi_t(S^{1/(1-\eta)}) - M + p_t(1-\eta)\chi M^\eta$ is jointly concave in S, M . The transformed first-order and costate conditions follow by the chain rule from the original equations. Moreover, $pS = qB/(1-\eta)$, so the original developer TVC implies $D_T p_T S_T \rightarrow 0$. Applying the supporting-hyperplane argument in these coordinates yields

$$\tilde{J}_T - J_T \leq -D_T p_T (\tilde{S}_T - S_T) \leq \frac{D_T q_T B_T}{1-\eta} \rightarrow 0.$$

Thus both ranges of η give global developer optimality against all admissible research paths. The argument applies to any candidate satisfying the uncapped equations and developer TVC under the proposition's restrictions, not just the BGP.

For the household, set $\Lambda_t = e^{-\rho t} N_t / C_t$. Euler implies $\dot{\Lambda} = -r\Lambda$. Concavity of log utility and the linear budget give, for any feasible alternative with the same K_0 and the same given prices and distributions,

$$\int_0^T e^{-\rho t} N_t \left[\log(\tilde{C}_t / N_t) - \log(C_t / N_t) \right] dt \leq -\Lambda_T (\tilde{K}_T - K_T) \leq \Lambda_T K_T \rightarrow 0.$$

This establishes global household optimality. Competitive final-good firms face a concave constant-returns technology, and Lemma 1 establishes global static monopoly optimality. The constructed path satisfies Definition 1 with the uncapped RSI technology (42).

Local convergence from nearby initial stocks. Use the ordering $\xi = (\xi_K, \xi_B, \xi_C, \xi_q)'$ in (141), and let e_K, e_B, e_C, e_q be the corresponding unit row vectors. Define the row vectors

$$y_\xi = \left(\frac{\alpha}{1 - (1 - \alpha)\omega_X}, \frac{(1 - \alpha)\omega_X}{1 - (1 - \alpha)\omega_X}, 0, 0 \right), \quad (121)$$

$$m_\xi = \frac{\eta e_B + e_q}{1 - \eta}, \quad (122)$$

$$b_\xi = g_B^*[(\eta - 1)e_B + \eta m_\xi]. \quad (123)$$

The reduced production equation gives the log deviation of output as $y_\xi \xi$. The research FOC implies $M = (q\chi\eta B^\eta)^{1/(1-\eta)}$, so the log deviation of M is $m_\xi \xi$. Differentiating $g_B = \chi B^{\eta-1} M^\eta$ at the BGP gives the level deviation $b_\xi \xi$.

The four rows of J^* , derived from capital accumulation, AI research, the household Euler equation, and the developer costate equation, are

$$J_K^* = \frac{1 - u^*}{k^*} y_\xi - (g_Y^* + \delta) e_K - \frac{c^*}{k^*} e_C - \frac{m^*}{k^*} m_\xi, \quad (124)$$

$$J_B^* = b_\xi, \quad (125)$$

$$J_C^* = (r^* + \delta)(y_\xi - e_K), \quad (126)$$

$$J_q^* = (r^* + \delta)(y_\xi - e_K) - (\rho - n + \eta g_Y^*)(y_\xi - e_q - e_B) - \eta b_\xi. \quad (127)$$

The last row uses $U/(qB) = \rho - n + \eta g_Y^*$ at the BGP. All entries depend on growth rates and ratios, which are independent of χ . No terminal restrictions were used to derive this Jacobian.

I next establish the two local properties (142) and (143) from the parameter restrictions, rather than impose them. For this linearization only, define

$$\begin{aligned} \nu &= \frac{1 - \eta}{\eta}, & a_K &= \frac{\alpha}{1 - (1 - \alpha)\omega_X}, & a_T &= \frac{(1 - \alpha)\omega_X \eta}{[1 - (1 - \alpha)\omega_X](1 - \eta)}, \\ d &= \rho - n, & P &= r^* + \delta, & h &= \rho - n + \eta g_Y^*. \end{aligned}$$

All six quantities are positive, and the restriction $\eta + \omega_X < 1$ gives

$$1 - a_K - a_T = \frac{(1 - \alpha)(1 - \eta - \omega_X)}{[1 - (1 - \alpha)\omega_X](1 - \eta)} > 0. \quad (128)$$

Make the invertible linear change of deviations

$$\widehat{\xi} = (\xi_K, \nu\xi_B, \xi_C, \xi_q + (1 - \nu)\xi_B)'$$

This changes neither the model nor the eigenvalues. Applying it to (124)–(127) gives a similar matrix

$$\widehat{J} = \begin{pmatrix} A & B & -C & -V \\ 0 & 0 & 0 & D \\ -G & H & 0 & 0 \\ -E & F & 0 & d \end{pmatrix}, \quad (129)$$

where the following letters abbreviate matrix coefficients only:

$$\begin{aligned} A &= d + (1 - \alpha)\omega_X P, & B &= \frac{(1 - u^*)a_T - m^*}{k^*}, & C &= \frac{c^*}{k^*}, \\ V &= \frac{m^*}{k^*(1 - \eta)}, & D &= \frac{\eta g_Y^*}{1 - \eta}, & G &= P(1 - a_K), \\ H &= Pa_T, & E &= G + ha_K, & F &= H + h(1 - a_T). \end{aligned}$$

All coefficients other than B are positive by their definitions. The BGP allocation ratios imply the identities

$$L \equiv B - \frac{Vd}{D} = \frac{(1 - \alpha)\omega_X \eta}{(1 - \eta)k^*} > 0, \quad (130)$$

$$GF - HE = hP(1 - a_K - a_T) > 0. \quad (131)$$

Set $N = CG + VE > 0$ and $Q = CH + VF > 0$. Expanding the determinant gives the characteristic polynomial

$$\begin{aligned} p(\ell) &= \det(\ell I - \widehat{J}) \\ &= (\ell^2 - A\ell - N)(\ell^2 - d\ell - DF) - DE(Q - L\ell). \end{aligned} \quad (132)$$

In particular,

$$p(0) = DChP(1 - a_K - a_T) > 0.$$

Let $\mu = (d - \sqrt{d^2 + 4DF})/2 < 0$, the negative root of the second quadratic factor. Then $p(\mu) = -DE(Q - L\mu) < 0$. Since $p(\ell) \rightarrow +\infty$ as $\ell \rightarrow -\infty$, there is one negative root below μ and another between μ and zero. These are the only negative roots: the cubic coefficient of p is $-(A + d) < 0$, its linear coefficient is $ADF + dN + LDE > 0$, and its constant coefficient is positive. Thus the coefficient signs of $p(-x)$, in descending powers, are $(+, +, \text{arbitrary}, -, +)$. Descartes' rule of signs

gives exactly two sign changes, regardless of the quadratic coefficient. The two negative roots are therefore real, distinct, and simple. Write them as $\ell_1 < \mu < \ell_2 < 0$. The remaining two roots have positive product and positive sum, because $p(0) > 0$ and the sum of all four roots is $A + d > 0$. If real, both are positive; if complex, their real parts are positive. This proves (142).

To establish the state projection, write a stable eigenvector in transformed coordinates as $(x, y, z, u)'$. Its second component cannot be zero: the second, fourth, and first rows would successively imply $u = x = z = 0$. Normalize $y = 1$. The second and fourth rows then give

$$x(\ell) = \frac{DF + d\ell - \ell^2}{DE}.$$

The ordering of the roots implies $x(\ell_1) < 0 < x(\ell_2)$. Hence the two projected stable eigenvectors $(x(\ell_1), 1)'$ and $(x(\ell_2), 1)'$ are linearly independent. Returning to (ξ_K, ξ_B) only rescales their second component by $1/\nu > 0$, so (143) follows as well.

The exact normalized vector field is continuously differentiable near $\xi = 0$. The spectral property just proved and the stable-manifold theorem supply a two-dimensional local manifold whose trajectories remain near zero and converge exponentially. The derivative of its projection onto (ξ_K, ξ_B) is V_{KB} . The nonsingularity just proved and the inverse-function theorem therefore make this manifold locally the graph of a continuously differentiable function selecting (ξ_C, ξ_q) from the two initial stock deviations. Since exponentiation is a local diffeomorphism, the same holds for (C_0, q_0) as functions of (K_0, B_0) . More explicitly, for some constants $a, H > 0$, trajectories on a sufficiently small stable manifold satisfy

$$\|\xi(t)\| \leq H e^{-at} \|\xi(0)\|, \quad t \geq 0.$$

For any sufficiently small open neighborhood $\mathcal{V}$ of zero, choose a ball contained in $\mathcal{V}$. Continuity of the graph selecting the two jump variables permits an open neighborhood $\mathcal{U}$ of the initial stock deviations for which $H\|\xi(0)\|$ is smaller than that ball's radius. Every initial stock pair in $\mathcal{U}$ therefore has a trajectory staying in $\mathcal{V}$ for all $t \geq 0$ and converging to zero. Exponentiation maps $\mathcal{U}$ to an open neighborhood of (K_0^*, B_0^*) . Shrinking these neighborhoods preserves positivity, feasibility, and existence for every $t \geq 0$. Local uniqueness follows because every trajectory remaining in this local region and converging to zero belongs to the same stable manifold, which is a graph over stocks. This does not assert convergence from arbitrary initial C_0, q_0 or exclude equilibrium trajectories that leave the region.

The statement also permits nearby A_0, N_0 . Equations (113) and (114) make the reference B_0^*, K_0^* smooth functions of these two positive initial values. After normalization by the corresponding BGP, the exact vector field depends only on the parameters and BGP ratios, not on A_0, N_0 . The set of (A_0, N_0, K_0, B_0) whose stock deviations belong to $\mathcal{U}$ is therefore open and contains the reference initial condition. From each such initial condition, the construction converges to the BGP associated with its own A_0, N_0 , not necessarily to the same levels as the original reference path.

Since $f(\xi(t)) \rightarrow 0$, growth rates converge to their BGP values, not only normalized levels. The

static equations give the same convergence for the interest rate, wage growth, and allocation ratios. The unit-elastic labor income share remains $(1 - \alpha)\omega_L$ at every date.

Exponential convergence implies that $r_t - r^*$ is integrable. Thus D_t/e^{-r^*t} tends to a finite positive constant, and both TVC expressions equal their BGP counterparts in (116) and (117) times factors tending to finite positive limits. Both TVCs hold. Output and all expenditure flows are bounded multiples of $e^{g_Y^*t}$, while log consumption per person is a linear function of time plus a bounded deviation. Both objective integrals therefore remain finite. The global concavity verification established above applies at these new price and allocation paths, proving agent optimality. Static optimality and market clearing then establish equilibrium, not merely a solution of the necessary conditions. $\square$

B Numerical algorithm and equilibrium audit

B.1 Economic intuition and initial conditions

The BVP selects an equilibrium trajectory, not an initial-value solution from arbitrary consumption and shadow prices. Capital and AI efficiency are predetermined; consumption and the developer's shadow value adjust to put the economy on the stable continuation of its long-run regime. An initial choice that satisfies the static conditions alone need not do so.

The RSI-activation exercises first solve the same auxiliary economy: AI is already used in production, but its efficiency is fixed because RSI is unavailable. For each σ , a scalar root search in $\log K_0$ solves Equation (61) using the original production and monopoly equations. Equation (62) then gives pre-event consumption. Homogeneity extends these date-zero values to a pre-event BGP: K, Y, C, U, X grow at $n + \gamma$, the wage grows at γ , and B, p_X, r and income shares are constant. Since $r - (n + \gamma) = \rho - n > 0$, the developer's pre-event profit has finite present value. The household's transversality expression decays at $n - \rho < 0$. Research is unavailable, rather than suppressed despite a profitable research opportunity.

At the unanticipated activation, only the states are inherited. The post-event problem uses the original positive- χ equilibrium equations; it does not impose pre-event consumption or a post-event zero-research corner. A separate audit checks continuity of the production quantities and prices at date zero and the pre-event Euler, resource, and monopoly conditions. It also records the endogenous consumption and research jumps.

B.2 BVP and calibration

I retain the collocation solver `scipy.integrate.solve_bvp` and the computational coordinates

$$\left(\log K, \log \frac{B}{\bar{B} - B}, \log C, \log q \right).$$

Subtracting the analytical terminal aggregate growth rate times t removes their linear trends without changing any model variable or its timing. Effective labor remains exogenous. At every

trial date and state, the code solves the static monopoly problem and recovers research from its first-order condition. The left boundary fixes K_0, B_0 ; two right-boundary projections remove the unstable components of the normalized equilibrium dynamics near the long-run limit. The regime-specific normalization and stable continuation are those of Appendix A.

For each elasticity, I start near its analytical limiting point and move the initial stocks to their prescribed values in 32 continuation stages. This is a numerical construction inside the positive-AI, positive-RSI problem; neither $\omega_X = 0$ nor $\chi = 0$ is used as a degenerate BVP seed. At unit elasticity the static block uses its exact Cobb–Douglas limit; near one, `log1p` and `expm1` avoid cancellation.

The principal calibration in Section 6.2 uses an approximate research-expenditure calibration at $\sigma = 1$. For each trial χ , Gauss–Legendre quadrature with 64 and 128 nodes evaluates the two integrals in Equation (63); their ratio must agree within 10^{-8} in logs. I reuse the increasing low- χ branch from the earlier numerical search and select its candidate closest to the 2023 proxy. The selected $\chi = 1.4378$ produces an annual ratio of 0.06374%, below the 0.06638% target by 0.26 basis points. This is an approximate empirical match, not a root solved to numerical precision. Publication requires a discrepancy below one basis point as well as all the original equilibrium checks. The annual moment must also remain stable within 2×10^{-5} in logs after both horizon extensions. I hold this unit-elastic calibration fixed for $\sigma = 0.90, 1.10, 1.50$ and solve each equilibrium from its own fixed-efficiency pre-RSI BGP. Research expenditure and the price decline in these three cases are outcomes, not additional targets. The earlier high-target search and its figures remain available in the repository.

For the sensitivity in Section 6.3, an outer bracketed search in $\log \chi$ solves the unit-elastic BVP and evaluates the price ratio at exactly 2.25 years. It matches Equation (64), then fixes χ for the other three elasticities. The tolerance in $\log \chi$ is 2×10^{-6} ; the final price-ratio log error must be below 2×10^{-5} after both horizon extensions. These numerical tolerances are well below the precision of the rounded target. The archived full-price-decline exercise uses the same method with a price ratio of 0.20 instead of 0.60.

B.3 Equilibrium checks and replication

I extend each computational horizon twice by 500 years and tighten the collocation tolerance from 2×10^{-6} to 10^{-8} and 10^{-9} . The final boundary tolerance is 10^{-11} . The initial meshes have 221, 401, and 601 points and adapt as needed. The displayed window lies within each solved horizon; no spline is extrapolated.

The original equations are reconstructed independently with five-point differences at 1,001 dates and steps of 0.003 and 0.001 years. An additional 801-date check over the first ten years uses steps of 0.0003 and 0.0001. Admission requires dynamic residuals below 10^{-6} , first-order residuals below 10^{-9} , final-horizon changes below 2×10^{-5} in detrended logs, and terminal-coordinate gaps below 10^{-4} . These are unchanged from the earlier algorithm; none relaxes an economic condition.

The audit also checks feasibility and the developer’s global optimality. When global profit concavity in research depth fails, I apply the Hamiltonian support condition of Lemma 2, with

counterfactual-state minimization and an analytical bound beyond the grid. Two grids use 81/101 and 321/241 date/state points, supplemented by dense first-decade checks. The analytical terminal support bound and the asymptotic rate $n - \rho < 0$ of both transversality expressions support the infinite-horizon continuation. These are numerically verified equilibrium approximations, not interval-arithmetic existence certificates for arbitrary initial stocks.

Table 4: Numerical accuracy of the principal research-expenditure calibration

σ	Final horizon	Mesh points	Dynamic residual	Horizon change
0.90	6,063.3	836	9.96×10^{-10}	5.01×10^{-8}
1.00	5,579.8	888	1.06×10^{-9}	3.74×10^{-8}
1.10	5,464.1	903	1.17×10^{-9}	4.00×10^{-8}
1.50	3,648.3	1,100	1.21×10^{-9}	1.87×10^{-8}

Dynamic residuals are the largest independently reconstructed errors across the full-horizon and dense first-decade checks. Horizon change compares the last two solutions in detrended logarithmic coordinates over years 0–500. Error bounds are rounded upward. The unit-elastic initial candidate uses the stricter tolerance 2×10^{-7} , followed by 10^{-8} and 10^{-9} . Event-continuity log gaps are below 1.34×10^{-15} . The unit-elastic annual expenditure ratio is 0.000637427339 after both horizon extensions; its log change from the initial candidate is below 3.44×10^{-8} . Across all four elasticities that change is below 6.47×10^{-7} . The remaining difference from the empirical target is a calibration discrepancy, not a numerical error or a relaxation of equilibrium conditions. The $\sigma = 1.50$ case passes both Hamiltonian-support grids and the analytical continuation bound; the other three cases pass the global concavity check.

Table 5: Numerical accuracy with the 40% price-decline target

σ	Final horizon	Mesh points	Dynamic residual	Horizon change
0.90	5,422.1	1,199	1.20×10^{-9}	3.09×10^{-8}
1.00	4,938.6	1,295	1.38×10^{-9}	3.50×10^{-8}
1.10	4,822.8	1,336	1.45×10^{-9}	3.74×10^{-8}
1.50	3,007.1	1,589	1.81×10^{-9}	1.50×10^{-8}

Definitions and admission tolerances are those of Table 4. Error bounds are rounded upward. The event-continuity log gaps are below 1.4×10^{-15} . After both horizon extensions, the unit-elastic price ratio is 0.6000000059.

From the repository root, reproduce the principal comparison with:

```
python scripts/calibrate_rewrite_research_share_low.py
```

This command reproduces the selected unit-elastic calibration and all four equilibrium paths, repeats the equilibrium and activation checks, and exports the three comparison figures. The optional argument `--sigma 1.5`, for example, solves only that case; the command without arguments requires all four cases to pass before publishing. The [research-expenditure results folder](#) contains the sources, admitted CSV, numerical checks, figure metadata and approximate-calibration record.

For the faster-transition sensitivity, run:

```
python scripts/calibrate_rewrite_ai_price.py --variant rsi_activation_half_decline
```

Its outputs are in the [half-decline results folder](#). Each comparison has separate checkpoints, calibration results, audits and figures. No paths or figures are exported unless every scenario passes

admission and the activation continuity check. Checkpoint and CSV hashes bind the audits to the plotted data. The [replication guide](#) gives installation instructions and the staged workflow; `scripts/reproduce_rewrite_results.py` runs both displayed comparisons.

The full 80%-price-decline exercise is preserved in the [full-price-decline results folder](#). To reproduce it, use the price calibration command with `--variant rsi_activation` instead. Instructions for the former high-expenditure-target exercise remain in the [preserved 2025-target results folder](#).

Earlier simulations and their numerical appendix remain in the source. The replication guide explains how to reactivate the full-price benchmark or restore the older quantitative section and numerical appendix using the two switches in `main_rewrite.tex`. These switches change the presentation, not any numerical file.

C The uncapped economy with unit elasticity

This appendix derives the rates and allocation ratios supporting Proposition 4 in Section 5.2 and defines the normalized variables used to prove local convergence. I use the uncapped RSI technology (42) and the production function (51). In this appendix and its proofs, a superscript $*$ denotes the reference BGP characterized in Section 5.2, distinguishing it from nearby equilibrium trajectories.

C.1 An explicit balanced-growth equilibrium

At unit elasticity, $s_X = \omega_X$ and $e_X = 1 - (1 - \alpha)\omega_X$. The AI-demand and monopoly conditions, equations (13) and (19), give

$$\frac{p_X X}{Y} = (1 - \alpha)\omega_X, \quad \frac{U}{Y} = (1 - \alpha)^2 \omega_X^2. \quad (133)$$

Substituting $X = BU = (1 - \alpha)^2 \omega_X^2 BY$ into production yields

$$Y^{1-(1-\alpha)\omega_X} = K^\alpha (AL)^{(1-\alpha)\omega_L} [(1 - \alpha)^2 \omega_X^2 B]^{(1-\alpha)\omega_X}. \quad (134)$$

On an exact balanced-growth path with positive constant K/Y , C/Y , and M/Y , log differentiation of (134) and (42) gives the two growth restrictions (52) and (53). Define

$$\Delta \equiv (1 - \alpha)(1 - \eta - \omega_X). \quad (135)$$

For $\Delta > 0$, their unique solution has

$$g_Y^* = \frac{(1 - \eta)(1 - \omega_X)}{1 - \eta - \omega_X} (n + \gamma), \quad (136)$$

and g_B^* as in (58). The equality $g_K = g_C = g_Y^*$ follows from constant capital and consumption ratios; equations (12) and (5) then give (55) and (56).

Define the candidate ratios $k^* = K/Y$, $u^* = U/Y$, $i^* = (\dot{K} + \delta K)/Y$, $m^* = M/Y$, and

$c^* = C/Y$ by

$$k^* = \frac{\alpha}{r^* + \delta}, \quad u^* = (1 - \alpha)^2 \omega_X^2, \quad (137)$$

$$i^* = (g_Y^* + \delta)k^*, \quad (138)$$

$$m^* = \frac{u^* \eta g_B^*}{\rho - n + \eta g_Y^*}, \quad (139)$$

$$c^* = 1 - u^* - i^* - m^*. \quad (140)$$

In this appendix, k^* is a capital–output ratio, not capital per unit of effective labor. The research share comes from the developer’s first-order and costate conditions; the consumption share comes from market clearing. These ratios are endogenous, not fixed saving or research rules. Along the BGP, $B_t = B_0^* e^{g_B^* t} \rightarrow \infty$, $g_K = g_C = g_U = g_M = g_Y^*$, $g_X = g_Y^* + g_B^*$, and $g_q = g_Y^* - g_B^*$.

The conditions have different roles. Since $n + \gamma > 0$, equations (52) and (53) require $\Delta > 0$ for this positive finite-rate BGP. If $\Delta = 0$, they are inconsistent; if $\Delta < 0$, they imply $g_B < 0$, contrary to the RSI technology. Thus $\eta + \omega_X \geq 1$ excludes this BGP, not every possible equilibrium trajectory. When $\eta \leq 1/2$, the bound $(1 - \alpha)\omega_X \leq 1/2$ makes operating profit concave in B , and the RSI technology is jointly concave in B, M . When $\eta > 1/2$, the proof instead uses the state $S = B^{1-\eta}$. The restrictions $\eta < \alpha$ and $\eta + \omega_X < 1$ then ensure concavity in the transformed problem. Thus $\eta < \alpha$ supports this second verification argument, although the BGP growth identities do not require it separately. None of these concavity tests is asserted to be necessary for equilibrium existence. Finally, $\rho > n$ makes both transversality expressions decline and the objective integrals converge.

C.2 Equilibrium trajectories near the balanced-growth path

The BGP supplies a reference for proving existence from nearby initial stocks that need not be on that path. Define stationary log deviations

$$\xi_K = \log \frac{K}{K^*}, \quad \xi_B = \log \frac{B}{B^*}, \quad \xi_C = \log \frac{C}{C^*}, \quad \xi_q = \log \frac{q}{q^*}. \quad (141)$$

Substitution into the four equilibrium dynamics gives a smooth autonomous system $\dot{\xi} = f(\xi)$ with $f(0) = 0$, where $\xi = (\xi_K, \xi_B, \xi_C, \xi_q)'$. Let $J^* = Df(0)$ denote its Jacobian. The proof in Appendix A derives its entries directly from equations (29a)–(29d), with $\psi = 1$ and $\psi' = 0$.

The parameter restrictions in Proposition 4 imply two local properties. First, the linearized dynamics have exactly two stable directions and no eigenvalue with zero real part. Writing E^s for the real invariant subspace associated with eigenvalues having negative real parts, the first property is

$$\text{spec}(J^*) \cap i\mathbb{R} = \emptyset, \quad \dim E^s = 2. \quad (142)$$

Second, these stable directions allow independent changes in the two predetermined stocks. Let

$V_S \in \mathbb{R}^{4 \times 2}$ be any real basis matrix for E^s , and let V_{KB} consist of its first two rows. Then

$$\det V_{KB} \neq 0. \quad (143)$$

Nonsingularity does not depend on the choice of basis. It allows the stable manifold to select C_0, q_0 for every sufficiently close pair K_0, B_0 , rather than only for stocks on a curve. Both properties follow analytically from the proposition's parameter restrictions; neither is an additional assumption or a calibration-specific test. The complete argument, including both transversality conditions and global agent optimality, is in Appendix A, [Proof of Proposition 4](#).

No constant growth rates or allocation shares are imposed along these transitions. The result is local in initial stocks, not in time. In particular, it concerns equilibrium choices of consumption and research, not convergence from arbitrary nearby values of all four variables: there are also two unstable directions. It does not rule out equilibria that leave the local neighborhood. The normalized Jacobian is independent of χ , A_0 , and N_0 ; changing research productivity changes the reference levels, not these roots.

C.3 Growth and distribution without an upper bound

The uncapped BGP has

$$\frac{wL}{Y} = (1 - \alpha)\omega_L > 0, \quad \frac{(r + \delta)K}{Y} = \alpha, \quad \frac{\Pi}{Y} = (1 - \alpha)\omega_X - u^* - m^*. \quad (144)$$

Equation (55) shows that real wages and output per person grow together at a rate above γ , even though labor retains a positive, constant income share. Without an upper bound, continued growth in AI efficiency sustains this wage-growth premium. With any finite upper bound, the unit-elastic equilibria in Proposition 1(ii) instead have $g_{Y/N}, g_w \rightarrow \gamma$. The equivalence between wage growth above γ and a vanishing labor share across the regimes with a finite upper bound does not extend to the uncapped economy.

Within the restrictions of Proposition 4, increasing the AI weight raises the common growth rate and the interest rate:

$$\frac{\partial g_{Y/N}^*}{\partial \omega_X} = \frac{\partial g_w^*}{\partial \omega_X} = \frac{\partial r^*}{\partial \omega_X} = \frac{\eta(1 - \eta)(n + \gamma)}{(1 - \eta - \omega_X)^2} > 0. \quad (145)$$

Labor's income share falls across these parameterizations, but remains constant over time on each BGP. As $\omega_X \rightarrow 0^+$, the common per-person growth rate converges to γ and $r^* \rightarrow \rho + \gamma$. These statements compare balanced-growth equilibria; they do not describe a transition from a no-AI economy or establish an uncapped equilibrium for $\sigma > 1$.

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